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Grade	**0.00** out of 169.00 (**0**%)

Question **1**

Not answered

Marked out of 1.00

Mr. Liam Gideon is a 43-year-old man who has recently been diagnosed with Huntington's disease after starting to exhibit characteristic movement abnormalities and subtle cognitive and mood changes. Which of the following is characteristic of the pathology of Huntington's disease?

Select one:

- Neuronal loss in the substantia nigra
- Neuronal loss and gliosis in the caudate nucleus and the putamen
- Relative sparing of the hippocampus
- Shrinking of the lateral ventricles
- Astrocytosis

Low voltage beta activity
Hypsarrhythmia
High voltage alpha activity
3–4 Hz spike and slow wave complexes
Progressive slowing of background rhythms
2Hz spike and slow wave complexes
Polyspike and wave discharge

Your answer is incorrect.

Early neuronal loss in the caudate and putamen of the basal ganglia along with associated gliosis are characteristic of Huntington's disease. Neuronal loss in the substantia nigra is more characteristic of Parkinson's disease but can occur in Huntington's disease. Atrophy seen in Huntington's does not spare the hippocampus and eventually results in ventriculomegaly. Astrocytosis usually occurs but this is non-specific and is a response to neuronal loss regardless of cause.

The correct answer is: Neuronal loss and gliosis in the caudate nucleus and the putamen

Question **2**

Not answered

Marked out of 4.00

Flag question

Theme- EEG findings
Lead In: Which of the above are seen on EEGs in the below disorders?

1. Delirium tremens
2. Juvenile myoclonic epilepsy
3. Absence seizures
4. West syndrome

Explanation: 1) Delirium tremens, unlike other delirium states which are characterised by diffuse slowing of background rhythms to alpha, theta and finally delta waves, tends to show excessive but low voltage fast wave, or beta, activity of 13-30Hz.
2) Juvenile myoclonic epilepsy is associated with characteristic 4-6Hz polyspike and wave discharges. Other abnormalities can also be seen, including generalized single spike waves.
3) In typical absence seizures the ictal EEG shows a characteristic regular and symmetrical generalised discharges of 3-4 Hz "spike and slow" wave complexes, and may also have multiple spike and slow wave complexes.
4) West Syndrome, or infantile spasms, is a severe form of epilepsy, usually presenting in early adulthood, which is associated with hypsarrhythmia on EEG, which is chaotic, high amplitude slow, sharp and poly-spike waves.
Ref: Britton JW, Frey LC, Hopp JLet al., authors; St. Louis EK, Frey LC, editors. Electroencephalography (EEG): An Introductory Text and Atlas of Normal and Abnormal Findings in Adults, Children, and Infants [Internet]. Chicago: American Epilepsy Society; 2016. EEG in the Epilepsies. Available from: https://www.ncbi.nlm.nih.gov/books/NBK390347/

Question **3**

Not answered

Marked out of 1.00

Flag question

Which of the following brain regions, when impaired in its function, severely limits empathetic expression during social judgments?

Select one:

- Prefrontal cortex
- Orbitofrontal cortex
- Amygdala
- Supramarginal gyrus
- Caudate nucleus

Your answer is incorrect.

The supramarginal gyrus, particularly on the right, appears to be most associated with empathy. Ref: G. Silani, C. Lamm, C. C. Ruff, T. Singer. Right Supramarginal Gyrus Is Crucial to Overcome Emotional Egocentricity Bias in Social Judgments. Journal of Neuroscience, 2013; 33 (39): 15466

The correct answer is: Supramarginal gyrus

Question **4**

Not answered

Marked out of 1.00

Flag question

A large population of neurons with which one of the following neurotransmitters is located within the locus coeruleus?

Select one:

- Dopamine
- Noradrenaline
- Acetylcholine
- Serotonin
- Oxytocin

Your answer is incorrect.

The locus coeruleus is a small nucleus in the dorsal pons comprising the main source of noradrenaline in the brain via its diffuse projections. It has projections through the brain and into the spinal cord.

The correct answer is: Noradrenaline

Question **5**

Not answered

Marked out of 1.00

Flag question

Mr Philip Davies is a 75-year-old man who experiences a stroke affecting his left angular gyrus. Which of the following abnormalities would be most likely to occur as a result of this?

Select one:

- Broca's
- Anomia
- Wernicke's
- Conduction
- Alexia

Your answer is incorrect.

Damage to the left angular gyrus is associated with alexia and Gerstmann syndrome, characterised by dysgraphia/agraphia, dyscalculia/acalculia, finger agnosia and left-right disorientation.

The correct answer is: Alexia

Question **6**

Not answered

Marked out of 1.00

Flag question

Anti-NMDA encephalitis is an autoimmune disease causing inflammation of the brain associated with acute psychotic symptoms. Which class of immunoglobulin in the CSF should be tested when suspecting anti-NMDA encephalitis?

Select one:

- IgM
- IgE
- IgG
- IgD
- IgA

Your answer is incorrect.

NMDAR antibody test is most sensitive and specific with CSF. Serum test for NMDAR antibody has a low false positive rate but a higher false negative rate. Pathogenic NMDAR autoantibodies are IgG responses. Both IgM and IgA responses are non-specific and are seen in patients with schizophrenia and other psychiatric disease as well as ~10% of normal controls. As a result, we cannot use the IgM and IgA responses for diagnosing autoimmune encephalitis. Serum or CSF testing for IgG antibodies against the NR1a subunit of the NMDA receptor is typically performed when autoimmune encephalitis is suspected. NMDAR IgG responses seen in anti-NMDAR encephalitis are not seen in schizophrenia. Ref: Lancaster E. The diagnosis and treatment of autoimmune encephalitis. Journal of Clinical

Neurology. 2016 Jan 1;12(1):13.

The correct answer is: IgG

Question **7**

Not answered

Marked out of 1.00

Flag question

Which area of the brain is affected first in terms of atrophy (volume loss) seen in Alzheimer's disease?

Select one:

- Prefrontal cortex
- Hippocampus
- Entorhinal cortex
- Paracentral lobule
- Amygdala

Your answer is incorrect.

MRI studies have demonstrated that the entorhinal cortex is affected first in Alzheimer's disease. It is an area of the brain located in the medial temporal lobe and functions as the main interface between the hippocampus and neocortex. (Ref: Khan UA, Liu L, Provenzano FA, Berman DE, Profaci CP, Sloan R, Mayeux R, Duff KE, Small SA (2013). Molecular drivers and cortical spread of lateral entorhinal cortex dysfunction in preclinical Alzheimer's disease. Nature Neuroscience. 17: 304–311. doi:10.1038/nn.3606. PMC 4044925. PMID 24362760)

The correct answer is: Entorhinal cortex

Question **8**

Not answered

Marked out of 1.00

Flag question

Orexin (or hypocretin) is a hormone produced in the hypothalamus and is associated with a variety of bodily functions. Which of the following conditions is most associated with low levels or orexin?

Select one:

- Depression
- Narcolepsy
- Parkinson's dementia
- Generalised anxiety disorder
- REM sleep disorder

Your answer is incorrect.

The most common form of narcolepsy, in which the sufferer experiences brief losses of muscle tone (cataplexy), is caused by a lack of orexin in the brain due to destruction of the cells that produce it. Low levels of orexin have been associated with feelings of sadness, while high levels are associated with happiness. High levels of orexin have been associated with increased appetite, adipose storage and addiction behaviours in animal studies.

The correct answer is: Narcolepsy

Question **9**

There are numerous psychological tests available for various cognitive functions. Which of the

Not answered

Marked out of 1.00

Flag question

following is a test of abstract reasoning?

Select one:

- Figure copying
- Clock drawing
- Number span
- Proverb testing
- Serial subtraction

Your answer is incorrect.

Proverb testing involves individuals interpreting common idioms, such as "a rolling stone gathers no moss" which could suggest that people who are always moving, with no roots in one place or another, avoid responsibilities and cares. If an individual can only provide reasoning using concrete methods, they would struggle to make these interpretations.

The correct answer is: Proverb testing

Question **10**

Not answered

Marked out of 1.00

Flag question

Mrs. Jane Francis is a 75-year-old woman who has presented acutely to her local A&E after developing sudden onset paralysis of her right arm and leg, as well as a loss of vision on her right side. A lesion in which of the following arteries is most likely to be the cause of her symptoms?

Select one:

- Circle of Willis
- Left anterior cerebral artery
- Left middle cerebral artery
- Right anterior cerebral artery
- Right middle cerebral artery

Your answer is incorrect.

This would be most likely due to a lesion affecting the left middle cerebral artery causing contralateral paralysis and homonymous hemianopia.

The correct answer is: Left middle cerebral artery

Question **11**

Not answered

Marked out of 1.00

Flag question

Neurotransmitter receptors in the brain function in a variety of ways. Some are G protein coupled and so pass through the cell membrane 7 times and detect molecules outside the cell and activate internal signal transduction pathways and, ultimately, cellular responses. Which of the following is a G protein coupled receptor?

Select one:

- AMPA
- 5-HT3
- NMDA

- GABA-A
- 5-HT1

Your answer is incorrect.

Most neurotransmitter receptors listed in the question are G-protein coupled, with the exception of 5-HT3. , 5HT3 receptor is a ligand-gated ion channel (ionotropic receptor), selectively permeable for cations such as sodium, potassium and calcium. Thus it has an excitatory function. Ref: Thompson AJ, R Lummis SC. 5-HT3 receptors. Current pharmaceutical design. 2006 Oct 1;12(28):3615-30.

The correct answer is: 5-HT1

Question **12**

Not answered

Marked out of 1.00

Flag question

The dorsal corpus striatum is an important structure within the basal ganglia. It is comprised of the putamen and

Select one:

- Olfactory tubercle
- Nucleus accumbens
- Caudate nucleus
- Substantia nigra
- Subthalamic nucleus

Your answer is incorrect.

The dorsal striatum is made up of the caudate nucleus and the lentiform nucleus. The lentiform nucleus is made up of the larger putamen, and the smaller globus pallidus. The ventral striatum is composed of the nucleus accumbens and olfactory tubercle.

The correct answer is: Caudate nucleus

Question **13**

Not answered

Marked out of 1.00

Flag question

Mr. Percy Coombes is a 68-year-old man who has suffered bilateral amygdala damage. Which of the following clinical feature can be expected?

Select one:

- Emotionality
- Anterograde amnesia
- Absence of fear response
- Hyposexuality
- Anosmia

Your answer is incorrect.

Bilateral amygdala damage is associated with an absence of fear responses to external stimuli. Interesting, fear responses can still be evoked by inhalation of high concentrations of CO2. More extensive bilateral damage to the area around the amygdala can cause Kluver-Bucy syndrome.

The correct answer is: Absence of fear response

Question **14**

Not answered

Marked out of 1.00

Flag question

Mr Jamie Hughes is a 25-year-old man with persistent difficulties with impulsive and aggressive behaviour. Which of the following abnormalities would be most likely to be identified in his CSF?

Select one:

- Low 5-HIAA
- High 5HT-3
- High 5HT-1
- Low 5HT-3
- High 5-HIAA

Your answer is incorrect.

Low levels of CSF 5-HIAA (5-hydroxyindole-acetic acid) have been consistently identified as being associated with aggressive and impulsive behaviour across a variety of studies and cultures.

The correct answer is: Low 5-HIAA

Question **15**

Not answered

Marked out of 1.00

Flag question

The negative symptoms of schizophrenia are often viewed as being at least as disabling as the positive ones, if not more so. Which of the following dopamine pathways is implicated in negative symptoms?

Select one:

- Tuberoinfundibular
- Mesolimbic
- Nigrostriatal
- Hypothalamospinal
- Mesocortical

Your answer is incorrect.

The mesolimbic dopamine pathway dysfunction is associated with positive symptoms, whereas the mesocortical dysfunction is associated with negative symptoms. It is important to note that this is a simplistic reduction that does not hold true in experimental models.

The correct answer is: Mesocortical

Question **16**

Not answered

Marked out of 1.00

Flag question

Huntington's disease is a genetic condition associated with characteristic movement abnormalities, and affective and cognitive dysfunction. Which of the following is the mutation seen in Huntington's disease?

Select one:

- CCG trinucleotide repeat

- CTG trinucleotide repeat
- AGG trinucleotide repeat
- CGG trinucleotide repeat
- CAG trinucleotide repeat

Your answer is incorrect.

Huntington's disease occurs due to a CAG trinucleotide repeat at chromosome 4p16.

The correct answer is: CAG trinucleotide repeat

Question **17**

Not answered

Marked out of 1.00

Flag question

Trinucleotide repeat mutations cause a variety of clinical conditions, including Huntington's disease and Fragile X syndrome. How would you test for a DNA trinucleotide repeat mutation?

Select one:

- Northern blot
- Western blot
- Eastern blot
- Southern blot
- Southwestern blot

Your answer is incorrect.

A Southern blot is a method used in molecular biology for detection of a specific DNA sequence in DNA samples. Northern blot is for RNA, Western for proteins, Eastern for post-translational modifications (including the addition of lipids, phosphates, and glycoconjugates), Southwestern for DNA-binding proteins, and Northwestern is a hybrid method for detecting interactions between RNA and proteins.

The correct answer is: Southern blot

Question **18**

Not answered

Marked out of 1.00

Flag question

Mr. Mark Fredericks is a 28-year-old man with a diagnosis of PTSD associated with prominent sleep disturbance. To investigate this further he undergoes sleep studies. Which of the following would be the most likely finding?

Select one:

- Decreased stage 1 sleep
- Decreased REM sleep latency
- Increased REM sleep latency
- Greater REM density
- Reduced stage 3 and 4 sleep

Your answer is incorrect.

Sleep studies in PTSD have been somewhat equivocal, although a meta-analysis of polysomnographic studies conducted with military veterans and civilian adults with PTSD found modest indices of objective sleep disruption in PTSD, as indicated by more stage 1 sleep, less slow-wave sleep and greater REM density than in subjects without PTSD. (Ref: Kobayashi I, Boarts JM, Delahanty DL. Polysomnographically measured sleep abnormalities in PTSD: a meta-analytic review. Psychophysiology. 2007;44:660–669).

The correct answer is: Greater REM density

Question **19**

Not answered

Marked out of 1.00

Flag question

Parkinson's disease is associated with particular pathological changes within the brain, giving rise to the characteristic movement abnormalities and later cognitive changes. In which of the following areas of the brain are Lewy Bodies most commonly found in Parkinson's disease?

Select one:

- Putamen
- Subthalamic nucleus
- Substantia nigra
- Globus pallidum
- Caudate nucleus

Your answer is incorrect.

Lewy Bodies are most prominent in the substantia nigra in Parkinson's disease. The main pathological characteristics of Parkinson's disease are cell death in the brain's basal ganglia (affecting up to 70% of the dopamine secreting neurons in the substantia nigra pars compacta by the end of life) and the presence of Lewy bodies in many of the remaining neurons. This loss of neurons is accompanied by the death of astrocytes and a significant increase in the number of microglia in the substantia nigra.

The correct answer is: Substantia nigra

Question **20**

Not answered

Marked out of 1.00

Flag question

Mr. Alan Bradley is a 66-year-old man who has been admitted with a sudden onset of quadriplegia and incontinence after a head injury. He is unable to speak but can communicate by blinking. Which of the following brain areas is most likely to be damaged?

Select one:

- Basal ganglia
- Hypothalamus
- Pons
- Cerebellum
- Midbrain

Your answer is incorrect.

This describes locked-in syndrome which can be classified into 3 categories: in the classic form patients are quadriplegic and anarthric with preserved consciousness and upper eyelid and vertical eye movements; in the incomplete form, patients still have limited voluntary movements other than vertical

eye movements, most frequently of fingers, toes and head; in the total form, patients show complete immobility including eye movements but have preserved consciousness. It is usually due to traumatic or vascular damage to the brain stem, vertebral or basilar artery damage or occlusion, or compression of the cerebral peduncles from tentorial herniation. It is most often caused by damage to a specific part of the brainstem known as the pons. In locked-in syndrome there is an interruption of all the motor fibers running from grey matter in the brain via the spinal cord to the body's muscles and also damage to the centers in the brainstem important for facial control and speaking. Ref: Laureys S, et al. The locked-in syndrome: what is it like to be conscious but paralyzed and voiceless?. Progress in brain research. 2005 Jan 1;150:495-611.

The correct answer is: Pons

Question **21**

Not answered

Marked out of 1.00

Flag question

Mrs. Eva Clark is a 72-year-old woman who has experienced a stroke affecting her dominant parietal lobe. Which of the following is she most likely to experience as a result?

Select one:

- Anosognosia
- Constructional apraxia
- Dyscalculia
- Spatial orientation
- Dressing apraxia

Your answer is incorrect.

Damage to the dominant parietal lobe is likely to lead to: dysphasia/aphasia, dyscalculia, dyslexia, apraxia, tactile agnosia and Gerstmann syndrome (acalculia, agraphia, finger agnosia and right-left disorientation). The other options presented are more likely to occur if the non-dominant parietal lobe is damaged.

The correct answer is: Dyscalculia

Question **22**

Not answered

Marked out of 1.00

Flag question

Mrs. Joanna Chapel is an 80-year-old woman who has attended a memory clinic and has been diagnosed with Alzheimer's disease. Her son is seeking information on the amyloid deposits that are supposedly accumulating in her brain. Which of the following pairs of enzymes is a precursor of the amyloid products seen in this dementing illness?

Select one:

- Beta and gamma secretase
- Alpha and gamma secretase
- Acetyl and butyrylcholinesterase
- Alpha and beta secretase
- Reelin and alpha secretase

Your answer is incorrect.

Accumulation of amyloid β-peptide (Aβ) is the main source of neuronal degeneration in Alzheimer's

disease (AD). When β-amyloid precursor protein (APP) is sequentially cleaved by β- and γ-secretase enzymes, this results in Aβ (amyloidogenic pathway). When α-secretase acts on APP, this precludes Aβ formation (non-amyloidogenic pathway). Reelin is a separate protein, which may interact with APP to have a role in Alzheimer's disease. Acetylcholinesterase is not a APP enzyme.

The correct answer is: Beta and gamma secretase

Question **23**

Not answered

Marked out of 1.00

Flag question

Betty Turner is a young girl who works hard in school, as she likes being praised by the teacher. Which of the following stages of development does this correspond to?

Select one:

- Industry vs. inferiority
- Ego identity vs. role confusion
- Autonomy vs. shame
- Initiative vs. guilt
- Trust vs. mistrust

Your answer is incorrect.

This is most likely to represent industry vs. inferiority in Erikson's stages of development, with this leading to a feeling of competence, usually around the ages of 6-11 years. During this stage, individuals compare themselves more with their peers and may place emphasis on external validation to demonstrate their abilities.

The correct answer is: Industry vs. inferiority

Question **24**

Not answered

Marked out of 1.00

Flag question

Dopamine has various functions within the brain, some of which are associated with certain neuronal pathways. Which of the following is sometimes referred to as the reward pathway?

Select one:

- Mesolimbic
- Hypothalamospinal
- Nigrostriatal
- Tuberoinfundibular
- Mesocortical

Your answer is incorrect.

The mesolimbic pathway is sometimes referred to as the reward pathway. It connects the ventral tegmental area in the midbrain, to the ventral striatum of the basal ganglia in the forebrain. The ventral striatum includes the nucleus accumbens and the olfactory tubercle. The release of dopamine from the mesolimbic pathway into the nucleus accumbens regulates incentive salience (i.e. motivation and desire for rewarding stimuli) and facilitates reinforcement and reward-related motor function learning; it may also play a role in the subjective perception of pleasure.

The correct answer is: Mesolimbic

Question **25**

Not answered

Marked out of 1.00

Flag question

Human DNA codes for proteins via codons of nucleotides, which correspond to certain amino acid. How many different codon combinations are seen in the human DNA?

Select one:

- 32
- 16
- 24
- 64
- 128

Your answer is incorrect.

DNA consists of four different bases, and because there are three bases in a codon, and 4 x4 x 4 = 64, there are 64 possible patterns for a codon.

The correct answer is: 64

Question **26**

Not answered

Marked out of 1.00

Flag question

Which of the following is an immediate precursor of dopamine during its synthesis?

Select one:

- Tryptophan
- P-Tyramine
- L-Tyrosine
- L-Dopa
- M-Tyramine

Your answer is incorrect.

There is one primary and two minor pathways for the synthesis of dopamine. These are:
• Primary: L-Phenylalanine → L-Tyrosine → L-DOPA → Dopamine
• Minor: L-Phenylalanine → L-Tyrosine → p-Tyramine → Dopamine
• Minor: L-Phenylalanine → m-Tyrosine → m-Tyramine → DopamineThere is one primary and two minor pathways for the synthesis of dopamine. These are:
• Primary: L-Phenylalanine → L-Tyrosine → L-DOPA → Dopamine
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• Minor: L-Phenylalanine → L-Tyrosine → p-Tyramine → Dopamine
• Minor: L-Phenylalanine → m-Tyrosine → m-Tyramine → Dopamine

The correct answer is: L-Dopa

Question **27**

Not answered

Marked out of 1.00

Flag question

Mr. James Crow is a 70-year-old man who has been diagnosed with Parkinson's disease after presenting with a classic pill rolling tremor and bradykinesia. Which of the following is most commonly seen in Parkinson's?

Select one:

- Decreased D2 receptor sensitivity in the basal ganglia
- Increased D2 receptor sensitivity in the basal ganglia
- Decreased D2 receptor availability in the basal ganglia
- Decreased D2 receptor occupancy in the cortex
- Increased D2 receptor occupancy in the cortex

Your answer is incorrect.

Parkinson's disease is associated with an increase in the presence of D2 postsynaptic receptors in the basal ganglia and consequent hypersensitivity to dopamine agonists.

The correct answer is: Increased D2 receptor sensitivity in the basal ganglia

Question **28**

Not answered

Marked out of 1.00

Flag question

Progranulin is a protein that is thought to have a role in brain development though its function is not fully understood yet. Which of the following diseases is progranulin associated with?

Select one:

- Supranuclear palsy
- Lewy Body dementia
- Vascular dementia
- Frontotemporal dementia
- Corticobasal degeneration

Your answer is incorrect.

Progranulin mutations are associated with frontotemporal dementia.

The correct answer is: Frontotemporal dementia

Question **29**

Not answered

The various areas of the human brain appear to develop at different rates. Which of the following brain areas is considered to be the last to mature?

Marked out of 1.00

Flag question

Select one:

- Cingulate cortex
- Prefrontal cortex
- Broca's area
- Premotor cortex
- Orbitofrontal cortex

Your answer is incorrect.

The prefrontal cortex continues to mature well into the late part of early adult life.

The correct answer is: Prefrontal cortex

Question **30**

Not answered

Marked out of 1.00

Flag question

Mrs. Lucia Climbing is a 32-year-old woman who has a long history of waking abruptly in the night, sitting bolt upright and screaming while appearing briefly unaware of where she is. She cannot recall anything about why she has woken. At what stage in her sleep cycle is she most likely to be waking?

Select one:

- REM sleep
- Stage 3
- Stage 4
- Stage 1
- Stage 2

Your answer is incorrect.

Night terrors are described as occurring in stage 4 sleep or the transition from stage 3 to stage 4.

The correct answer is: Stage 4

Question **31**

Not answered

Marked out of 1.00

Flag question

Memory is the retention of information over time and it appears that there are various memory systems, which have their own associated brain structures. Which of the following is used in driving a car?

Select one:

- Short-term memory
- Episodic memory
- Procedural memory
- Long-term memory
- Semantic memory

Your answer is incorrect.

Procedural memory is not based on the conscious recall of information, but on implicit learning. It can best be summarized as remembering how to do something, such as driving a car.

The correct answer is: Procedural memory

Question **32**

Not answered

Marked out of 1.00

Flag question

Normal aging is associated with a reduction in brain size, even in the absence of marked cognitive decline. Which of the following nuclei is least likely to degenerate in old age?

Select one:

- Cranial nerve
- Inferior nuclei
- Caudate
- Putamen
- Superior nuclei

Your answer is incorrect.

Aging results in progressive loss of neuronal cell bodies in many regions of the brain and spinal cord. Motor neuron loss in the spinal cord leads to sarcopenia – a progressive age-related reduction in muscle mass. Cranial nerve nuclei withstand age-related degenerative changes to a large extent, though dendritic loss has been documented in several nuclei. Ref: Schwarz EC et al. The effects of aging on hypoglossal motoneurons in rats. Dysphagia. 2009 Mar 1;24(1):40.

The correct answer is: Inferior nuclei

Question **33**

Not answered

Marked out of 1.00

Flag question

Neuronal functioning relies upon a number of molecules, which can be divided into primary and secondary messengers. Which of the following acts as a second messenger in the Central Nervous System?

Select one:

- Serotonin
- Magnesium ions
- Potassium ions
- Calcium ions
- Adrenaline

Your answer is incorrect.

Second messengers are intracellular signaling molecules released by the cell in response to exposure to extracellular signaling molecules—the first messengers. Second messengers trigger physiological changes at cellular level such as proliferation, differentiation, migration, survival, and apoptosis. Examples of second messenger molecules include cyclic AMP, cyclic GMP, inositol trisphosphate, diacylglycerol, and calcium. First messengers are extracellular factors, often hormones or neurotransmitters, such as adrenaline, growth hormone, and serotonin.

The correct answer is: Calcium ions

Question **34**

Not answered

Marked out of 1.00

Flag question

There are several important proteolytic enzymes in the brain that influence the development of certain neurological conditions. Which of the following enzymes preclude amyloid formation?

Select one:

- Beta secretase
- COMT
- Monoamine oxidase
- Alpha secretase
- Gamma secretase

Your answer is incorrect.

Excess amyloid β-peptide (Aβ) contributes to the neuronal degeneration in Alzheimer's disease. Aβ is a 38 -43 amino acid peptide derived from the β-amyloid precursor protein (APP) through sequential cleavages by beta- and gamma-secretase enzymes. The cleavage site for another APP processing enzyme, alpha-secretase lies within the Aβ sequence. Thus, the action of alpha secretase leads to non-amyloidogenic products and precludes Aβ formation. Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2889200/

The correct answer is: Alpha secretase

Question **35**

Not answered

Marked out of 1.00

Flag question

Prions are misfolded proteins with the ability to transmit their misfolded shape onto normal variants of the same protein. They characterise several fatal and transmissible neurodegenerative diseases in humans and many other animals. Which part of the brain is most associated with prion diseases?

Select one:

- Frontal lobe
- Cerebellum
- Temporal lobe
- Hypothalamus
- Pons

Your answer is incorrect.

While prion diseases have a variety of clinical presentations, early symptoms generally include prominent cerebellar symptoms. As the disease progresses, patients generally develop global neurological symptoms. Mutations related to Prion diseases can result in different regions of the brain being affected. For example, thalamus is affected preferentially in fatal familial insomnia. While the cellular mechanisms involved in targeting specific neuronal populations and neuron-to-neuron spreading of prions in the CNS remain elusive, in several prion diseases, the cerebellum is a preferential prion target (Ref: Ragagnin A, et al. Prion Proteins and Neuronal Death in the Cerebellum. In Prions-Some Physiological and Pathophysiological Aspects 2018 Nov 5. IntechOpen).

The correct answer is: Cerebellum

Question **36**

Not answered

Marked out of 1.00

Flag question

Parkinson's disease is a progressive neurological condition characterised by certain motor symptoms, including tremor. Which of the following is true regarding the type of tremor seen in Parkinson's disease?

Select one:

- Associated with abnormal posturing
- Absent at rest
- Increases when target is nearer
- Higher amplitude than cerebellar tremor
- 4-6Hz in frequency

Your answer is incorrect.

The tremor associated with Parkinson's disease is present at rest, usually starts on one side of the body, has a frequency of 4-6Hz and is relatively low amplitude. Intention tremor is less common in Parkinson's disease than in essential tremor, but can occur. Ref; Thenganatt MA, Louis ED. Distinguishing essential tremor from Parkinson's disease: bedside tests and laboratory evaluations. Expert review of neurotherapeutics. 2012 Jun 1;12(6):687-96.

The correct answer is: 4-6Hz in frequency

Question **37**

Not answered

Marked out of 1.00

Flag question

Tau proteins are important in the CNS, both in terms of normal neurological functioning but also in the development of certain diseases. Which of the following is true regarding normal tau protein?

Select one:

- Stabilises microtubules
- Coded for an chromosome 15
- Mainly found in non-neurones in the CNS
- It does not interact with other proteins
- Phosphorylation results in dysfunction

Your answer is incorrect.

Tau proteins aid in the stabilisation of microtubules when they are functioning normally. They are present in higher concentrations in neurons and are coded on chromosome 17 by MAPT (microtubule-associated protein tau). Tau proteins interact with tubulin to stabilize microtubules and promote tubulin assembly into microtubules. Tau has two ways of controlling microtubule stability: isoforms and phosphorylation. Hyperphosphorylation results in Tau becoming dysfunctional and accumulating, leading to neuronal degeneration. Ref: Kadavath et al. Tau stabilizes microtubules by binding at the interface between tubulin heterodimers. Proceedings of the National Academy of Sciences. 2015 Jun 16;112(24):7501-6.

The correct answer is: Stabilises microtubules

Question **38**

Not answered

Which of the following diseases is Progranulin associated with?

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Flag question

Select one:

- Parkinson's disease
- Lewy body dementia
- Frontotemporal dementia
- Supranuclear palsy
- Alzheimer's disease

Your answer is incorrect.

Frontotemporal dementia is familial in 25% - 50% of cases (autosomal dominant with high penetrance). Genes causing this type of FTD include MAPT, the gene encoding tau, CHMP2B, and GRN, the gene that encodes progranulin. GRN mutation results in ubiquitinated TAR DNA binding protein (TARDBP or TDP-43) deposits while tau tangles are rare. Ref: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2901991/

The correct answer is: Frontotemporal dementia

Question **39**

Not answered

Marked out of 1.00

Flag question

Mr Gregory Perry is a 24 year old who is taken to A&E by his family after developing gradually increasing disorientation and confusion over several weeks and is now presenting with aggression and ill-formed delusions about people harming him. Due to him having some soft neurological symptoms, he is tested for autoimmune encephalitis. If this was the diagnosis, which of the following immunoglobulins would be most likely to be involved?

Select one:

- IgE
- IgD
- IgA
- IgM
- IgG

Your answer is incorrect.

Autoantibody testing is extremely important for the proper diagnosis of autoimmune encephalitis. Psychiatric manifestations are common early in the course of autoimmune encephalitis. These may include psychosis, aggression, inappropriate sexual behaviors, panic attacks, compulsive behaviors, euphoria or fear. Symptoms may fluctuate rapidly. Although this presentation is well known for anti-NMDAR encephalitis, anti-AMPAR and anti-GABA-B-R are also reported.. NMDAR and other cell surface antibody tests are most sensitive and specific with CSF. Serum may offer a low false positive rate and a higher false negative rate. Pathogenic cell surface or synaptic autoantibodies are IgG responses. NMDAR IgM and IgA responses have been reported in patients with schizophrenia and other psychiatric disease but also in up to 10% of normal controls; these IgM and IgA responses have no established role in diagnosing autoimmune encephalitis. (Ref: Lancaster E. The Diagnosis and Treatment of Autoimmune Encephalitis. J Clin Neurol. 2016;12(1):1–13. doi:10.3988/jcn.2016.12.1.1)

The correct answer is: IgG

Question **40**

Mr Andrews Martins is a 45-year-old man who has been involved in a major motorcycle accident

Not answered

Marked out of 1.00

Flag question

leaving him with residual cognitive changes. The neuropsychologist working with him is concerned about his executive function. Which of the following tests would be the appropriate to test his executive function?

Select one:

- Wechsler Scale
- Rey–Osterrieth complex figure
- Tower of London Test
- Boston Naming Test
- Bender-Gestalt Test

Your answer is incorrect.

The Tower of London test is a test used in applied clinical neuropsychology for the assessment of executive functioning specifically to detect deficits in planning.

The correct answer is: Tower of London Test

Question **41**

Not answered

Marked out of 1.00

Flag question

Mrs. Jane Simons is a 45-year-old woman who develops gradual onset low mood and cognitive slowing which is followed by gait disturbance, in-coordination and then abnormal movements. She is later diagnosed with Huntington's disease and she and her family undergo genetic counselling. Which of the following chromosomes is affected in Huntington's disease?

Select one:

- 12
- 4
- 1
- 6
- 10

Your answer is incorrect.

The Huntington gene is located on the short arm of chromosome 4. It contains a CAG trinucleotide repeat and if this exceeds 36 repeats, the individual is at risk of developing Huntington's disease, with penetrance being complete if there are 40 repeats or more.

The correct answer is: 4

Question **42**

Not answered

Marked out of 1.00

Flag question

The amygdalae are almond-shaped clusters of nuclei located deep and medially within the temporal lobes of the brain and are considered part of the limbic system. Which of the following is a primary function of the amygdala?

Select one:

- Emotional consolidation of salient memories
- Working memory

- Encoding long term memory
- Facial recognition
- Motivation

Your answer is incorrect.

The amygdalae have been shown to perform a primary role in the processing of memory, decision-making and emotional responses. They appear to regulate memory consolidation in other brain regions. During the consolidation period, the memory can be modulated. In particular, it appears that emotional arousal following the learning event influences the strength of the subsequent memory for that event. Greater emotional arousal following a learning event enhances a person's retention of that event. Ref: https://www.ncbi.nlm.nih.gov/pubmed/15217324

The correct answer is: Emotional consolidation of salient memories

Question **43**

Not answered

Marked out of 1.00

Flag question

Suicide prevention is an increasingly important area of academic study. Which of the following neurochemical abnormalities is associated with an increased risk of suicide?

Select one:

- Increased 5-HIAA level in CSF
- Reduced noradrenaline in locus coeruleus
- Decreased GABA-A responsiveness
- Increased GABA-A responsiveness
- Decreased 5-HIAA level in CSF

Your answer is incorrect.

Reduced 5-HIAA levels (5-hydroxy indole acetic acid) in the CSF have been associated with aggression and suicide by violent means. They are thought to be related to reduced serotonin levels in the brain, as 5-HIAA is a metabolite of serotonin. Ref: Mehlman PT, et al. Low CSF 5-HIAA concentrations and severe aggression and impaired impulse control in nonhuman primates. The American journal of psychiatry. 1994 Oct.

The correct answer is: Decreased 5-HIAA level in CSF

Question **44**

Not answered

Marked out of 1.00

Flag question

Metabotropic receptors are indirectly linked to ion channels via signal transduction mechanisms, such as G-proteins. Which of the following is a metabotropic receptor?

Select one:

- AMPA
- Glycine
- Endocannabinoid
- Kainate
- GABA-A

Your answer is incorrect.

Metabotropic receptors include muscarinic acetylcholine, metabotropic glutamate, GABA-B receptors as well as most serotonin receptors. The endocannabinoid CB(1) or CB(2) receptors are also metabotropic, though the endocannabinoids anandamide has been observed to have activating properties at certain voltage gated calcium ion channels. Ref; Di Marzo V. New approaches and challenges to targeting the endocannabinoid system. Nature Reviews Drug Discovery. 2018 Sep;17(9):623-39.

The correct answer is: Endocannabinoid

Question **45**

Not answered

Marked out of 1.00

Flag question

Cocaine is a common drug of abuse that has significant addictive potential. Which of the following brain areas is particularly affected by cocaine?

Select one:

- Nucleus accumbens
- Corpus callosum
- Left anterior parietal lobe
- Parahippocampal gyrus
- Cerebellum

Your answer is incorrect.

The nucleus accumbens is particularly associated with cocaine misuse as it has a key role in mediating reward mechanisms within the brain and influences long-term dependence. Cocaine activates numerous areas within the brain during acute use, particularly within the limbic system.

The correct answer is: Nucleus accumbens

Question **46**

Not answered

Marked out of 1.00

Flag question

Miss Charlotte Rankin is a 25-year-old woman who has to attend an alcohol awareness course after being convicted of drink-driving. She is surprised to learn that, on an average, alcohol is only metabolised at the rate of around 1 unit per hour, regardless of how much is in the body. Which of the following is the main rate-limiting step in alcohol metabolism?

Select one:

- Acetyl-CoA
- Catalase
- Aldehyde dehydrogenase
- Cytochrome P450
- Alcohol dehydrogenase

Your answer is incorrect.

The enzyme alcohol dehydrogenase activity is a major rate limiting factor for ethanol metabolism in humans; as a result, excess consumption of alcohol induces zero order kinetics wherein only a fixed amount of alcohol gets oxidised. Ref: Jones AW. Alcohol, its absorption, distribution, metabolism, and

excretion in the body and pharmacokinetic calculations. Wiley Interdisciplinary Reviews: Forensic Science. 2019 Sep;1(5):e1340.

The correct answer is: Alcohol dehydrogenase

Question **47**

Not answered

Marked out of 1.00

Flag question

The dementias all have characteristic pathological changes on macroscopic and microscopic examination of brain tissue. Which of the following is characterised by microscopic spherical bodies?

Select one:

- Alzheimer's disease
- Lewy body dementia
- Pick's disease
- Vascular dementia
- CJD

Your answer is incorrect.

Pick bodies are the characteristic morphology assumed by tau filaments in Pick's disease, in which they accumulate in limbic (especially hippocampal dentate gyrus) and cortical neurons. They are intraneuronal, approximately the size of the nucleus and are formed of disorganized bundles of filaments comprised of three 3-repeat isoforms of tau. They appear as silver-staining, spherical-shaped aggregations.

The correct answer is: Pick's disease

Question **48**

Not answered

Marked out of 1.00

Flag question

Emma George is a 26-year-old woman who is experiencing repeated night mares. In which of the following stages of sleep are nightmares most likely to occur?

Select one:

- REM sleep
- Stage 4
- Stage 2
- Stage 1
- Stage 3

Your answer is incorrect.

Nightmares occur in REM sleep.

The correct answer is: REM sleep

Question **49**

Not answered

Marked out of

Transcranial magnetic stimulation (rTMS) is a novel treatment in which a changing magnetic field is used to cause electric current at a specific area of the brain through electromagnetic induction. It has been studied in a variety of neurological and psychiatric conditions. Which of the following lobes of the brain is targeted when it is used in treatment resistant depression?

1.00

Flag question

Select one:

- Left frontal
- Right parietal
- Left temporal
- Right temporal
- Left parietal

Your answer is incorrect.

For treatment-resistant major depressive disorder, targeting of the left dorsolateral prefrontal cortex (DLPFC) appears effective and low-frequency (LF) rTMS of the right DLPFC has probable efficacy. The most promising areas to target for OCD appear to be the orbitofrontal cortex.

The correct answer is: Left frontal

Question **50**

Not answered

Marked out of 1.00

Flag question

Visualising specific tracts in the brain is useful in research and certain clinical circumstances. Which of the following is the best imaging technique for identifying a tract specific lesion?

Select one:

- fMRI
- DTI
- DAT
- CT
- MRI

Your answer is incorrect.

Diffusion tensor imaging (DTI) is a magnetic resonance imaging technique that enables the measurement of the restricted diffusion of water in tissue in order to produce neural tract images instead of using this data solely for the purpose of assigning contrast or colors to pixels in a cross sectional image. The principal direction of the diffusion tensor can be used to infer the white matter connectivity of the brain. (This process is called tractography). For example, see Palaniyappan L, et al. Neuropsychopharmacology. 2013 Aug;38(9):1808.

The correct answer is: DTI

Question **51**

Not answered

Marked out of 1.00

Flag question

The pituitary gland, or hypophysis, is an endocrine gland about the size of a pea. It is a protrusion off the bottom of the hypothalamus at the base of the brain. It produces numerous hormones which influence a wide variety of bodily processes. Which of the following hormones are not synthesised in the pituitary?

Select one:

- Prolactin
- Vasopressin

- Growth hormone
- Leptin
- Melanocyte–stimulating hormone

Your answer is incorrect.

Leptin is generally produced in the adipose tissue and enterocytes of the small intestine. The anterior pituitary produces: growth hormone, adrenocorticotropic hormone (ACTH), beta-endorphin, melanocyte-stimulating hormone, thyroid-stimulating hormone, luteinizing hormone, follicle-stimulating hormone and prolactin. The intermediate lobe produces melanocyte-stimulating hormone. The posterior pituitary produces: vasopressin (anti-diuretic hormone) and oxytocin. Ref; Harris RB. Biochimica et Biophysica Acta (BBA)-Molecular Basis of Disease. 2014 Mar 1;1842(3):414-23.

The correct answer is: Leptin

Question **52**

Not answered

Marked out of 1.00

Flag question

Mr Lawrence Graves is a 75 year old man with a diagnosis of probable Alzheimer's disease which is currently mild in severity. He is wondering about what is happening in his brain to cause his memory impairment. Which of the following is thought to be the earliest histological change in Alzheimer disease?

Select one:

- Ubiquitin-mediated damage
- Tau-mediated injury
- Gliosis
- Lewy body deposition
- Amyloid-mediated damage

Your answer is incorrect.

The current hypothesis is that Alzheimer's disease starts with abnormalities in amyloid and the deposition of abnormal plaques. Some authors have suggested that this may be incorrect and the first change may be mediated by tau. The amyloidogenic theory is the most widely accepted at present.

The correct answer is: Amyloid-mediated damage

Question **53**

Not answered

Marked out of 1.00

Flag question

Schizophrenia is a major research topic within psychiatry, in terms of causation, treatment and prognosis. Which of the following is a recent focus of genetic research in schizophrenia that is associated with intellectual deficits?

Select one:

- Copy number variants
- Noradrenaline expression
- Dopamine receptor subtypes
- Heterogeneous genes
- Genetic heterogeneity

Your answer is incorrect.

A copy number variation (CNV) is when the number of copies of a particular gene varies from one individual to the next. They are recognised as contributing to both to normal genomic variability and risk for human diseases. Recent genome-wide studies have provided initial evidence that CNVs may also be associated with schizophrenia, especially in syndromes associated with intellectual deficits such as the Velocardiofacial Syndrome. Ref: Hiroi N, et al. Molecular psychiatry. 2013 Nov;18(11):1153.

The correct answer is: Copy number variants

Question **54**

Not answered

Marked out of 1.00

Flag question

Cortisol is a glucocorticoid, which is increased in response to stress and low blood-glucose concentration. It increases blood sugar through gluconeogenesis, suppresses the immune system, and aids in the metabolism of fat, protein, and carbohydrates. From which of the following areas is it secreted?

Select one:

- Adrenal gland
- Liver
- Kidney
- Hypothalamus
- Pituitary

Your answer is incorrect.

Cortisol is a steroid hormone, in the glucocorticoid class of hormones. It is produced in the zona fasciculata of the adrenal cortex within the adrenal gland.

The correct answer is: Adrenal gland

Question **55**

Not answered

Marked out of 1.00

Flag question

Leptin is a hormone that has a role in satiety and the development of obesity. From which of the following areas is it secreted?

Select one:

- Adrenal
- Adipose tissue
- Hypothalamus
- Liver
- Pituitary

Your answer is incorrect.

Leptin is a hormone predominantly made by adipose cells and enterocytes in the small intestine that helps to regulate energy balance by inhibiting hunger, which in turn diminishes fat storage in adipocytes.

The correct answer is: Adipose tissue

Question **56**

Not answered

Marked out of 1.00

Flag question

Insulin-like growth factor 1 (IGF-1), also called somatomedin C, is a hormone similar in molecular structure to insulin, which plays an important role in childhood growth, and has anabolic effects in adults. From which of the following areas is it secreted?

Select one:

- Kidney
- Liver
- Pituitary
- Adrenals
- Hypothalamus

Your answer is incorrect.

IGF-1 is produced primarily by the liver. Production is stimulated by growth hormone (GH). Ref: Laron Z. Insulin-like growth factor 1 (IGF-1): a growth hormone. Molecular Pathology. 2001 Oct;54(5):311.

The correct answer is: Liver

Question **57**

Not answered

Marked out of 1.00

Flag question

The Wechsler scale WAIS is a widely used neuropsychological test. Which of the following does it measure?

Select one:

- Intelligence
- Organisation
- Attention
- Visuospatial ability
- Executive function

Your answer is incorrect.

The Wechsler Adult Intelligence Scale (WAIS) is an IQ test designed to measure intelligence and cognitive ability in adults and older adolescents. There is an equivalent test for children. Note that there is also a Wechsler's memory scale (WMS).

The correct answer is: Intelligence

Question **58**

Not answered

Marked out of 1.00

Mr George Francis is a 22-year-old man who has been referred to his local neurology clinic by his GP after collapsing in the street and exhibiting some unusual movements. He is scheduled to have an EEG prior to a clinic appointment. Which of the following is true of EEGs in the investigation of possible seizure disorder?

Flag question

Select one:

- High negative predictive value
- High sensitivity
- High validity
- High specificity
- High positive predictive value

Your answer is incorrect.

Epileptiform activity is specific, but not sensitive, for diagnosis of epilepsy as the cause of a transient loss of consciousness or other paroxysmal event that is clinically likely to be epilepsy. EEG has relatively low sensitivity in epilepsy, ranging between 25–56%. Specificity is better, but again variable at 78–98%.

The correct answer is: High specificity

Question **59**

Not answered

Marked out of 1.00

Flag question

Mr Emile Sotoka is a 78 year old man who has presented to his GP with low mood, lack of enjoyment, difficulty focusing on activities and managing his daily tasks and diminished appetite. He has no previous history of significant health disorder. Which of the following would be likely to be identified on a CT scan?

Select one:

- Subcortical white matter opacities
- Periventricular white matter opacities
- Hippocampal atrophy
- Cerebellar atrophy
- Decreased ventricle size

Your answer is incorrect.

Elderly patients with depression are more likely to have greater cerebral sulcal atrophy; larger sylvian fissures, lateral ventricles, third ventricles, and temporal horns; and greater severity of subcortical white matter lesions. Depressed patients also exhibit more deep white matter hyperintensities but comparable levels of periventricular hyperintensity to controls. Ref; Salo KI et al. Frontiers in Psychology. 2019;10.

The correct answer is: Subcortical white matter opacities

Question **60**

Not answered

Marked out of 1.00

Flag question

Cocaine is a relatively common drug of abuse with significant additive potential. Which of the following neurotransmitters explain its effects?

Select one:

- Opioid
- Glutamate
- GABA

- Dopamine
- Glycine

Your answer is incorrect.

Cocaine acts by inhibiting the reuptake of serotonin, norepinephrine, and dopamine.

The correct answer is: Dopamine

Question **61**

Not answered

Marked out of 1.00

Flag question

Mrs Abigail Steuben is a 35-year-old woman with a diagnosis of treatment resistant schizophrenia who has been taking Clozapine for the past 12 months. She has now reported that she has not had a period for the past 4 months and found that she has a high prolactin level. She is not pregnant. Which of the following investigations would be most appropriate?

Select one:

- FSH/LH ratio
- MRI
- Growth hormone
- CT
- Visual fields testing

Your answer is incorrect.

Clozapine is an uncommon cause of hyperprolactinaemia. Alternative causal factors need to be investigated, particularly the possibility of a pituitary prolactinoma, which is best visualised by MRI.

The correct answer is: MRI

Question **62**

Not answered

Marked out of 1.00

Flag question

DOPA decarboxylase has an important role in the production of various neurotransmitters. Which of the following molecules is it involved in the synthesis of?

Select one:

- Adrenaline
- Melatonin
- Dopamine
- Noradrenaline
- Tyrosine

Your answer is incorrect.

DOPA decarboxylase, also called Aromatic L-amino acid decarboxylase, catalyses several different decarboxylation reactions:
• L-DOPA to dopamine – a neurotransmitter
• L-Phenylalanine to phenethylamine – a trace amine which functions as a neuromodulator
• L-Tyrosine to tyramine – a trace amine neuromodulator

• L-Histidine to histamine – a neurotransmitter
• L-Tryptophan to tryptamine – a trace amine neuromodulator
• 5-HTP to serotonin (5-hydroxytryptamine) – a neurotransmitter DOPA decarboxylase, also called Aromatic L-amino acid decarboxylase, catalyses several different decarboxylation reactions:
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• L-Tyrosine to tyramine – a trace amine neuromodulator
• L-Histidine to histamine – a neurotransmitter
• L-Tryptophan to tryptamine – a trace amine neuromodulator
• 5-HTP to serotonin (5-hydroxytryptamine) – a neurotransmitter

The correct answer is: Dopamine

Question **63**

Not answered

Marked out of 1.00

Flag question

The sleep-wake cycle is mediated by a wide range of brain areas. Which of the following areas are not involved in wakefulness?

Select one:

- Substantia nigra
- Cerebellum
- Dorsal raphe nucleus
- Basal forebrain
- Locus coeruleus

Your answer is incorrect.

A very wide variety of brain regions are associated with the sleep-wake cycle, with multiple interacting neurotransmitter systems in the brain stem, hypothalamus, and basal forebrain converging onto common effector systems in the thalamus and cortex.

The correct answer is: Cerebellum

Question **64**

Not answered

Marked out of 1.00

The pituitary gland is a key structure for homeostasis and is involved in the synthesis and secretion of a wide variety of hormones. It is divided into anterior, intermediate and posterior lobes. Which of the following is secreted by posterior pituitary lobe?

Select one:

Flag question

- Thyroid stimulating hormone
- Growth hormone
- Oxytocin
- Luteinizing hormone
- Melanocyte stimulating hormone

Your answer is incorrect.

The posterior pituitary produces: vasopressin (anti-diuretic hormone) and oxytocin. The anterior pituitary produces: growth hormone, adrenocorticotropic hormone (ACTH), beta-endorphin, melanocyte-stimulating hormone, thyroid-stimulating hormone, luteinizing hormone, follicle-stimulating hormone and prolactin. The intermediate lobe produces melanocyte-stimulating hormone.

The correct answer is: Oxytocin

Question **65**

Not answered

Marked out of 1.00

Flag question

Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence. Epigenetics most often involves changes that affect gene activity and expression, but the term can also be used to describe any heritable phenotypic change. Which of the following is not an epigenetic mechanism?

Select one:

- DNA acetylation
- DNA methylation
- Histone phosphorylation
- Histone methylation
- Glycation

Your answer is incorrect.

Glycation is the addition of a sugar molecule to a protein without the controlling action of an enzyme. The other mechanisms listed are known epigenetic processes, although there are many others, such as imprinting and alternative splicing.

The correct answer is: Glycation

Question **66**

Not answered

Marked out of 1.00

Flag question

Mr Brian Page is a 28-year-old man with a diagnosis of schizophrenia. He is currently well and asks his consultant about what causes his symptoms when unwell and how his brain may function differently to other people. Which of the following is most likely to be found in people with schizophrenia?

Select one:

- Decreased activity in prefrontal cortex
- Decreased cerebellar activity
- Decreased parietal lobe activity
- Increased temporal lobe activity

- Increased hippocampal activity

Your answer is incorrect.

Schizophrenia is associated with memory deficits and decreased hippocampal and dorsolateral prefrontal cortex activation.

The correct answer is: Decreased activity in prefrontal cortex

Question **67**

Not answered

Marked out of 1.00

Flag question

Short-term regulation of hunger and food intake involves neural signals from the GI tract, blood levels of nutrients, GI tract hormones, and psychological factors. Which of the following is involved in appetite and satiety?

Select one:

- Prolactin
- Growth hormone
- Cholecystokinin
- Oestrogen
- Follicle-stimulating hormone

Your answer is incorrect.

The hormones insulin and cholecystokinin are released from the GI tract during food absorption and act to suppress feelings of hunger. Cholecystokinin is key in suppressing hunger because of its role in inhibiting neuropeptide Y. Glucagon and adrenaline levels rise during fasting and stimulate hunger. Ghrelin, a hormone produced by the stomach, is a hunger stimulant. Leptin, a hormone produced by adipose tissue, acts as hunger suppressant.

The correct answer is: Cholecystokinin

Question **68**

Not answered

Marked out of 1.00

Flag question

The prefrontal cortex is associated with executive functioning, personality, language and planning. Which of the following best represents the percentage of the total brain volume made up by the prefrontal cortex?

Select one:

- 35
- 30
- 15
- 25
- 20

Your answer is incorrect.

The percentage of prefrontal cortex relative to total brain volume was 12.51 for humans and 10.68 for baboons. Similarly, the proportion of both dorsal and orbital prefrontal cortex volumes is larger in

human brains. Relative to total brain volume, the percentages of dorsal and orbital prefrontal cortex was 8.22% and 4.29% respectively in humans and 7.21% and 3.47% in baboons (Ref: McBride T, Arnold SE, Gur RC. A comparative volumetric analysis of the prefrontal cortex in human and baboon MRI. Brain, Behavior and Evolution; 1999 Sep;54(3):159-66).

The correct answer is: 15

Question **69**

Not answered

Marked out of 1.00

Flag question

Early onset Alzheimer's disease is associated with specific genetic variations. If an individual has two copies of the Apo E4 allele, how elevated is their risk of developing Alzheimer's disease?

Select one:

- 9x
- 12x
- 2x
- 3x
- 7x

Your answer is incorrect.

The presence of one Apo E4 allele has been associated with a 2-3 times in the risk of Alzheimer's disease, while the presence of two, is associated with an increase of around 8-15 times (median value of 12x). This varies by study and the effect is not as pronounced in all ethnic groups. (Ref: Williams T, et al. Therapeutic approaches targeting Apolipoprotein E function in Alzheimer's disease. Molecular Neurodegeneration. 2020 Dec 1;15(1):8).

The correct answer is: 12x

Question **70**

Not answered

Marked out of 1.00

Flag question

Huntington's disease is a genetic neurodegenerative disorder characterised by early cognitive changes, progressive motor symptoms and eventual dementia. An abnormality in which of the following genes causes it?

Select one:

- 19q11
- 1q31
- 9q69
- 4p16
- 14q73

Your answer is incorrect.

Huntington's disease is caused by an abnormally expanded CAG triplets repeat in the Huntingtin gene on the short arm of chromosome 4, at position 4p16.

The correct answer is: 4p16

Question **71**

Not answered

Marked out of 1.00

Flag question

Dementia with Lewy bodies and Parkinson's disease are characterised by particular clinical features and the accumulation of Lewy bodies within the brain. Which of the following molecules are contained within Lewy bodies?

Select one:

- Apolipoprotein
- Amyloid
- Ubiquitin
- Alpha synuclein
- Tau protein

Your answer is incorrect.

A classical Lewy body is an eosinophilic cytoplasmic inclusion and is primarily composed of alpha-synuclein.

The correct answer is: Alpha synuclein

Question **72**

Not answered

Marked out of 1.00

Flag question

Mr Louis Grovener is a 74 year old who has been referred to his local Older Person's Mental Health Service, as he has been experiencing difficulty naming objects and describing what they are for. His family has noticed that he has tended to be more vague in his speech, often saying "thing" or "things" rather than the names of common objects. Which of the following areas is most likely to be affected?

Select one:

- Prefrontal cortex
- Posterior parietal lobe
- Basal ganglia
- Anterior temporal lobe
- Medial temporal lobe

Your answer is incorrect.

This patient is likely to be suffering from semantic dementia or semantic variant primary progressive aphasia. It is one of the frontotemporal dementias. It is characterised by progressive anomia, fluent aphasia, impaired comprehension and visual agnosia in the early stages. This is due to an impairment of semantic memory which stores knowledge and facts. It preferentially affects the temporal lobes, with anterior areas and left lobe tending to show greater atrophy. (Ref: Hodges JR, Patterson K, Oxbury S and Funnell E. Semantic dementia. Progressive fluent aphasia with temporal lobe atrophy. Brain;1992 December;115 (6): 1783–1806).

The correct answer is: Anterior temporal lobe

Question **73**

Not answered

Marked out of 1.00

Mr. Laurence Walden is a 76-year-old man who has been having difficulty finding correct words to use during everyday conversations. He has attended his GP as this has started to significantly impact upon his daily life. His memory appears to be preserved. Which part of his brain is most likely to be affected?

Flag question

Select one:

- Wernicke's area
- Arcuate fasciculus
- Medial temporal lobe
- Medial frontal lobe
- Broca's area

Your answer is incorrect.

This scenario describes anomia, which is a form of aphasia characterised by difficulties in word retrieval, with individuals affected being unable to find and use the words they wish to. Similar difficulties can be caused by damage to Broca or Wernicke's area or to the angular gyrus. Anomia can also be a residual aspect of many aphasias, thus localizing it to a single region is considered difficult. But studies using diffusion imaging indicate that lesions affecting the left arcuate fasciculus, a white matter tract, are more likely to cause anomia. (Ref: Anderson JM, Gilmore R, Roper S, et al. Conduction aphasia and the arcuate fasciculus: A reexamination of the Wernicke-Geschwind model. Brain Language; 1999 October;70(1):1–12).

The correct answer is: Arcuate fasciculus

Question **74**

Not answered

Marked out of 1.00

Flag question

Dopamine is an important neurotransmitter involved in executive functioning, motor control, motivation and reward. Which of the following dopamine receptors are most abundant in the human brain?

Select one:

- D1
- D5
- D4
- D3
- D2

Your answer is incorrect.

The D1 receptor is the most abundant dopamine receptor in the brain, followed by D4. D1 receptor is linked to stimulatory G-proteins that activate adenylate cyclase. The D1 receptors are found in high concentration in the substantia nigra pars reticulata, caudate, putamen, nucleus accumbens, olfactory tubercle, and frontal and temporal cortex. (Ref: Martel JC, Gatti McArthur S. Dopamine Receptor Subtypes, Physiology and Pharmacology: New Ligands and Concepts in Schizophrenia. Frontiers in Pharmacology. 2020; 11:1003. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7379027/).

The correct answer is: D1

Question **75**

Not answered

Marked out of 1.00

Which one among the following imaging techniques measures the extent of myelination of white matter fibres at the macroscopic level?

Select one:

Flag question

- Functional MRI
- Arterial spin labelling
- Diffusion tensor imaging
- Resting state connectivity
- Single photon emission tomography

Your answer is incorrect.

Diffusion tensor imaging measures at the macroscopic level the extent of myelination of white matter fibres using fractional anisotropy, a measure of the degree and directionality of diffusion of water molecules. More recent advances in diffusion tensor imaging include deterministic and global probabilistic tractography, quantitative and more probabilistic measures respectively, of connectivity between reconstructed white matter fibres. Other techniques such as magnetization transfer imaging and quantitative T1 mapping can also be used to quantify the putative myelin content in the brain. (Ref: Palaniyappan L, et al. Aberrant myelination of the cingulum and Schneiderian delusions in schizophrenia: a 7T magnetization transfer study. Psychological medicine. 2019 Aug;49(11):1890-6.)

The correct answer is: Diffusion tensor imaging

Question **76**

Not answered

Marked out of 1.00

Flag question

ADHD is generally considered to be a lifelong condition that is thought to be heritable. Which of the following genes is associated with ADHD?

Select one:

- DRD2
- BDNF
- NOTCH3
- 5HT2C agonism
- DRD4

Your answer is incorrect.

The dopamine receptor D4 (DRD4) and D5 (DRD5) genes have been associated with ADHD. The dopamine transporter gene has also been associated with ADHD previously but evidence has been equivocal. (Ref:Palaniyappan Let al. Reduced Prefrontal Gyrification in Carriers of the Dopamine D4 Receptor 7-Repeat Allele With Attention Deficit/Hyperactivity Disorder: A Preliminary Report. Frontiers in Psychiatry. 2019 Apr 25;10:235.).

The correct answer is: DRD4

Question **77**

Not answered

Marked out of 1.00

Flag question

Which one among the following imaging technique examines changes in blood oxygen level dependent signal as a proxy measure of neural activity?

Select one:

- CT scan
- Functional MRI
- Structural MRI

- SPECT
- PET

Your answer is incorrect.

Functional MRI scan examines changes in blood oxygen level dependent signal (i.e. deoxygenation of haemoglobin) as a proxy measure of neural activity and is used to measure the functional integrity of neural activity.

The correct answer is: Functional MRI

Question **78**

Not answered

Marked out of 1.00

Flag question

Glutamate decarboxylase is an enzyme within the nervous system. It has a role in the synthesis of which of the following neurotransmitters?

Select one:

- Glycine
- Glutamate
- Serotonin
- GABA
- Dopamine

Your answer is incorrect.

Glutamate decarboxylase (GAD) catalyzes the decarboxylation of glutamate into GABA. GAD has two major isoforms, of 65 and 67 kDa, from two different genes. GAD67 is the major GABA synthesizing form; it is present throughout the soma, in the cytoplasm of inhibitory interneurons. GAD65 is enriched in axon terminals in linkage with synaptic vesicles. Ref: Melzer S, Monyer H. Diversity and function of corticopetal and corticofugal GABAergic projection neurons. Nature Reviews Neuroscience. 2020 Aug 3:1-7.

The correct answer is: GABA

Question **79**

Not answered

Marked out of 1.00

Flag question

GHB (gamma-Hydroxybutyric acid) is medically used as a general anaesthetic, but is also used recreationally. Which of the following receptors does it act on?

Select one:

- GABA-A
- Glutamate
- Glycine
- GABA-A and -B
- GABA-B

Your answer is incorrect.

GHB acts on the specific GHB receptors that are abundant in hippocampal dentate gyrus, median raphe nucleus as well as nucleus accumbens, mediating its effect on memory and sleep but also has weak agonistic action at the GABA-B receptor, mediating addictive properties and GABA-A receptor, mediating anxiolytic and hypnotic effects. Ref: Kamal RM, et al. The neurobiological mechanisms of gamma-hydroxybutyrate dependence and withdrawal and their clinical relevance: a review. Neuropsychobiology. 2016;73(2):65-80.

The correct answer is: GABA-B

Question **80**

Not answered

Marked out of 1.00

Flag question

There are various types of fronto-temporal dementia characterised by different pathological features. Which of the following is not a form of frontotemporal dementia?

Select one:

- Pick's disease
- Progressive non-fluent aphasia
- Progressive supranuclear palsy
- Huntington's disease
- Semantic dementia

Your answer is incorrect.

Huntington's disease typically affects the basal ganglia in the early phases of the disorder but then generally spreads to affect all cortical areas. Progressive supranuclear palsy was thought to be related to Parkinson's disease, but is now thought to be more related to frontotemporal dementia. (Ref: Finger EC. Frontotemporal Dementias. Continuum (Minneap Minn). 2016;22(2):464–489).

The correct answer is: Huntington's disease

Question **81**

Not answered

Marked out of 1.00

Flag question

Mr. Alan Marston is a 69-year-old man who has recently experienced a cerebrovascular accident. He is frustrated, as he cannot do things as usual. He cannot follow instructions about how to complete a task, such as using a hammer, but can perform each individual step. Which of the following best describes this phenomenon?

Select one:

- Constructional apraxia
- Oculomotor apraxia
- Ideomotor apraxia
- Ideational apraxia
- Limb-kinetic apraxia

Your answer is incorrect.

This describes ideational apraxia, which occurs when an individual loses the ability to plan movements related to using objects, as they appear to have lost the concept of what the item is for. Ideomotor apraxia is when an individual struggles to imitate or mime movements, such as brushing their hair or teeth. Oculomotor apraxia describes an individual being unable to move their eyes voluntarily; it is

seen in Balint's syndrome. Constructional apraxia is when an individual is unable to draw or copy shapes or patterns. Limb-kinetic apraxia is an impairment of co-ordinating voluntary movements of limbs.

The correct answer is: Ideational apraxia

Question **82**

Not answered

Marked out of 1.00

Flag question

Pregnancy is associated with numerous physiological changes, which can impact upon the effectiveness of medications. Which of the following occurs in pregnancy?

Select one:

- Decreased Glomerular Filtration Rate
- Increased gastric emptying
- Increase in gastric pH
- Decreased volume of distribution
- Increased protein binding of medication

Your answer is incorrect.

Pregnancy is associated with a decrease in the production of gastric acid, leading to an increase in overall gastric pH. Glomerular Filtration Rate (GFR) increases by 50% in the first trimester and continues to increase throughout pregnancy. Gastric emptying and intestinal motility is reduced during pregnancy. Plasma protein binding decreases over the course of pregnancy due to a drop in albumin levels. There are changes in hepatic metabolism of drugs, with some CYP isoforms showing increased activity, with others being less active. There is a marked increase in cardiac output and in the volume of distribution, both in terms of hydrophilic and lipophilic agents. Reference: Johnson-Davis KL, Doyle K. Therapeutic drug monitoring in pregnant patients. Therapeutic Drug Monitoring. 2020 Apr 1;42(2):172-80.



Pharmacokinetics of drugs in pregnant patients

Pharmacokinetic effect	Physiological change
Reduced drug absorption	Nausea & Vomiting
	Delayed gastric mobility
	Increased gastric pH (reduced acid secretion)
Increased volume of distribution	Increased body weight, body fat and plasma volume
Increased free drug concentration	Reduced plasma albumin
Increased drug clearance	Increased hepatic, renal blood flow
	Increased GFR

Johnson-Davis KL, Doyle K. Therapeutic drug monitoring in pregnant patients. Therapeutic Drug Monitoring. 2020 Apr 1;42(2):172-80.

SPMM COURSE

The correct answer is: Increase in gastric pH

Question **83**

Not answered

Marked out of 1.00

Flag question

Miss Elaine King is a 35-year-old woman with a diagnosis of paranoid schizophrenia. She experiences intermittent auditory hallucinations despite antipsychotic medication. Which of the following would you expect to see on functional imaging when she experiences these hallucinations?

Select one:

- Reduced activity in the posterior parietal cortex
- Increased activity in superior temporal lobe
- Reduced activity in the cingulate cortex
- Reduced activity in the hippocampus
- Increased activity in the amygdala

Your answer is incorrect.

Auditory hallucinations in schizophrenia have been associated with increased activity in a variety of brain areas including: inferior frontal gyrus/insula, middle temporal gyrus, superior temporal gyrus, middle frontal gyrus, posterior parietal cortex, thalamus, inferior colliculus, anterior cingulate gyrus, and left hippocampus and parahippocampal gyrus. Individuals with schizophrenia tend to have less grey matter than controls in the superior temporal lobe. (Ref: Jardri R, Sommer I. Functional brain imaging of hallucinations: symptom capture studies. InThe Neuroscience of Hallucinations 2013 (pp. 375-391). Springer, New York, NY).

The correct answer is: Increased activity in superior temporal lobe

Question **84**

Not answered

Marked out of 1.00

Flag question

Noradrenaline is a major neurotransmitter involved in arousal, alertness and anxiety. In which of the following brain areas is noradrenaline produced?

Select one:

- Cerebellar vermis
- Caudate nucleus
- Raphe nuclei
- Locus coeruleus
- Hypothalamus

Your answer is incorrect.

Noradrenaline is mostly produced in the locus coeruleus ('blue spot').

The correct answer is: Locus coeruleus

Question **85**

Not answered

Marked out of 1.00

Flag question

Mrs Charlotte Reynolds is a 75-year-old woman who has recently been diagnosed with mild Alzheimer's disease. Which of the following anatomical changes are most likely to be seen on a MRI scan of her brain?

Select one:

- Medial temporal lobe atrophy

- Enlargement of third ventricle
- Lateral ventricle enlargement
- White matter ischaemic changes
- Prefrontal cortex atrophy

Your answer is incorrect.

Medial temporal lobe, and particularly entorhinal, atrophy is considered to be typical of Alzheimer's disease and can be present prior to diagnosis. As the disease progresses, there is progressive neuronal loss across the cortex. (Ref: Jagust W. Imaging the evolution and pathophysiology of Alzheimer disease. Nature Reviews Neuroscience. 2018 Nov;19(11):687-700.).

The correct answer is: Medial temporal lobe atrophy

Question **86**

Not answered

Marked out of 1.00

Flag question

Recreational and therapeutic drugs can impact upon reward pathways in the brain, potentially resulting in dependence or increased reward-seeking behaviour. Which of the following is the reward pathway in the human brain?

Select one:

- Hypothalamospinal
- Mesolimbic
- Tuberoinfundibular
- Mesocortical
- Nigrostriatal

Your answer is incorrect.

The mesolimbic dopaminergic pathway is recognised as a major reward pathway in the brain, although other areas have a role, such as frontal areas. The mesocortical pathway is more associated with executive functions. The nigrostriatal pathway is associated with movement and associative learning. The tuberoinfundibular pathway is in the hypothalamus and its activation inhibits the release of prolactin. The hypothalamospinal tract influences motor function in the brainstem and spinal cord. (Ref: Maremmani I, Pacini M, Fountoulakis KN. Psychobiology of Addictions. In Psychobiology of Behaviour 2019 (pp. 171-192). Springer.).

The correct answer is: Mesolimbic

Question **87**

Not answered

Marked out of 1.00

Flag question

Recent studies on Lewy body dementia suggest hypoperfusion and hypometabolism in which part of the brain areas?

Select one:

- Prefrontal cortex
- Occipito-parietal
- Tempero-parietal
- Cerebellum

- Fronto-temporal

Your answer is incorrect.

Single photon emission computerised tomography (SPECT) and fluorodeoxyglucose positron emission tomography (PET) studies in dementia with Lewy bodies have demonstrated occipital and occipito-parietal visual association area hypoperfusion and hypometabolism respectively. However, occipital deficits are not consistently reported in dementia with Lewy bodies. Higher visual areas, particularly occipito-parietal, appear abnormal in dementia with Lewy bodies, while there is a preservation of function in lower visual areas (V1 and V2/3). (Ref: Ref: Yousaf T, Dervenoulas G, Valkimadi PE and Politis M. Neuroimaging in Lewy body dementia. Journal of Neurology; 2019 Jan;266(1):1-260)

The correct answer is: Occipito-parietal

Question **88**

Not answered

Marked out of 1.00

Flag question

Mrs. Olive Stanford is an 83-year-old woman who has attended her GP due to worries about her memory. It appears that her functioning is preserved and a MMSE score is normal. Her GP tells her that he feels her difficulties are likely to be part of normal ageing. Which of the following cognitive functions tend to be preserved during ageing?

Select one:

- Procedural memory
- Working memory
- Spatial awareness
- Retrograde memory
- Short term memory

Your answer is incorrect.

Procedural or non-declarative memory functions appear to be well preserved over the lifespan. Semantic memory or knowledge is also relatively preserved compared to episodic memory, although semantic knowledge often has a late life decline. Processing speed tends to decline in later life. (Ref: Lighthall, N. R., Conner, L. B., & Giovanello, K. S. (2019). Learning and memory in the aging brain: The function of declarative and nondeclarative memory over the lifespan. In G. R. Samanez-Larkin (Ed.), The aging brain: Functional adaptation across adulthood (p. 73–109). American Psychological Association.).

The correct answer is: Procedural memory

Question **89**

Not answered

Marked out of 1.00

Flag question

Structural imaging in which of the following types of dementias may reveal midbrain atrophy referred to as the hummingbird sign on sagittal images?

Select one:

- Huntington's disease
- Pick's disease
- Progressive supranuclear palsy
- Progressive non-fluent aphasia
-

Semantic dementia

Your answer is incorrect.

Structural imaging in PSP may reveal midbrain atrophy referred to as the hummingbird sign on sagittal images. In FDG-PET scans, hypometabolism in frontal, caudate, midbrain, and thalamic regions are notable in PSP. Excerpt from Finger EC. Frontotemporal Dementias. Continuum (Minneap Minn). 2016;22(2):464–489

The correct answer is: Progressive supranuclear palsy

Question **90**

Not answered

Marked out of 1.00

Flag question

Which of the following neurotransmitters is stored in the raphe nuclei?

Select one:

- GABA
- Glutamate
- Noradrenaline
- Serotonin
- Dopamine

Your answer is incorrect.

Serotonin is produced in large amounts in the neurons originating from the raphe nucleus in the brainstem, with wide-ranging projections to elsewhere in the nervous system. Noradrenaline is associated with the neurons of locus coeruleus in the pons.

The correct answer is: Serotonin

Question **91**

Not answered

Marked out of 1.00

Flag question

The frontal lobe is associated with a variety of neurological and cognitive functions, in terms of personality, executive functions and speech. Which of the following areas is not a part of the frontal cortex?

Select one:

- Primary motor cortex
- Broca's area
- Polar frontal cortex
- Secondary motor cortex
- Wernicke's area

Your answer is incorrect.

Wernicke's area is considered to be located in the temporal lobe, near temporo-parietal junction, on the surface of posterior portion of the superior temporal gyrus. The concept of a secondary motor area is variably described, but the premotor and supplementary motor cortices are located in the frontal lobe.

The correct answer is: Wernicke's area

Question **92**

Not answered

Marked out of 1.00

Flag question

The first pass effect is an important element of pharmacokinetics . Which of the following drugs has the least first pass effect?

Select one:

- Haloperidol
- Sulpiride
- Asenapine
- Diazepam
- Risperidone

Your answer is incorrect.

The first pass effect is the phenomenon whereby a proportion of an orally administered active drug is metabolised in the liver or gut before it reaches the systemic circulation. Sulpride and amisulpride both have low first pass effects in the liver and gut compared to many other psychotropics. They are extensively eliminated via the renal system. The other medications listed have high first pass effects.

The correct answer is: Sulpiride

Question **93**

Not answered

Marked out of 1.00

Flag question

Antidepressants can have variable effects of sleep, with some being described as activating and others sedating. Which of the following medications is most likely to cause insomnia?

Select one:

- Amitriptylline
- Trazodone
- Mirtazapine
- Agomelatine
- Duloxetine

Your answer is incorrect.

Duloxetine is associated with insomnia more often than the other medications listed here. SSRIs and venlafaxine are also associated with insomnia, and can be more activating than duloxetine in some cases.

The correct answer is: Duloxetine

Question **94**

Not answered

Marked out of 1.00

Flag question

Sodium valproate is known to increase the risk of congenital malformations if taken by pregnant women. What is the risk of an infant having a major congenital malformation if their mother takes valproate during pregnancy?

Select one:

- 41-50%

- 11-20%
- 21-30%
- 1-10%
- 31-40%

Your answer is incorrect.

Risk for any birth defect quoted 7.2% in Maudsley & NICE. North American registry quotes 10% risk. A large meta-analysis demonstrated that 10.73% of the children of women with epilepsy exposed to valproate monotherapy during pregnancy have congenital malformations. The risk appears to be dose dependent although there is no evidence that there is a dose below which no risk exists. (Ref: Drug Safety Update. MHRA. Medicines related to valproate:risk of abnormal pregnancy outcomes. Available at: www.gov.uk/drug-safety-update/medicines-related-to-valproate-risk-of-abnormal-pregnancy-outcomes)

The correct answer is: 1-10%

Question **95**

Not answered

Marked out of 1.00

Flag question

Sodium valproate is known to increase the risk of developmental delay and other cognitive difficulties in infants if it taken by their mother during pregnancy. What is the risk of this occurring?

Select one:

- 21-30%
- 11-20%
- 1-10%
- 31-40%
- 41-50%

Your answer is incorrect.

Around 30-40% of children exposed in utero to valproate show delays in early development, including walking and talking. (Ref: Drug Safety Update. MHRA. Medicines related to valproate:risk of abnormal pregnancy outcomes. Available at: www.gov.uk/drug-safety-update/medicines-related-to-valproate-risk-of-abnormal-pregnancy-outcomes)

The correct answer is: 31-40%

Question **96**

Not answered

Marked out of 1.00

Flag question

Tricyclic antidepressants are now used less frequently despite being very effective in treating depression. This is at least partially due to concerns regarding the issue of overdose. Which one of the following tricyclic antidepressants is the most dangerous in overdose?

Select one:

- Amitriptyline
- Imipramine
- Dosulepin(dothiepin)
- Clomipramine

- Lofepramine

Your answer is incorrect.

Both Amitriptyline and Dosulepin are recognised as being more dangerous in overdose, even when compared to other tricyclic antidepressants. Dosulepin is particularly dangerous and some mental health hospitals have removed it from their formularies.

The correct answer is: Dosulepin(dothiepin)

Question **97**

Not answered

Marked out of 1.00

Flag question

Pregabalin is being used more frequently in recent years. Which of the following is true of Pregabalin?

Select one:

- It passively diffuses across the blood-brain barrier.
- It is excreted without being metabolised by the liver
- It is effective in treating OCD
- It has an exceptionally long half life
- It is a GABA agonist

Your answer is incorrect.

Pregabalin experiences minimal metabolism in the body and is generally excreted unchanged in the urine. Its half-life is around 6 hours. Despite being a GABA analogue, it does not bind to GABA receptors, nor does it modulate GABA transport or metabolism. It is a ligand of certain voltage-dependent calcium channels, which it inhibits. It has low lipophilicity and is transported across the blood-brain barrier by LAT-1. Pregabalin's main indication is for generalised anxiety disorder.

The correct answer is: It is excreted without being metabolised by the liver

Question **98**

Not answered

Marked out of 1.00

Flag question

Miss. Amelie Toutou is a 27-year-old woman with a diagnosis of treatment resistant schizophrenia. When started on Clozapine she develops symptoms suggestive of myocarditis. Which of the following is a specific indicator of clozapine-induced myocarditis?

Select one:

- CRP
- ECG changes
- Chest x-ray
- Eosinophilia
- Troponin T

Your answer is incorrect.

Myocarditis is notoriously difficult to definitively diagnosis, with Troponin and Creatine Kinase levels often showing poor sensitivity for the disorder. In a review conducted of 47 cases of Clozapine induced myocarditis, the development of eosinophilia suggesting IgE mediated hypersensitivity

reaction has been documented in ~66% of cases, though interpreting isolated eosinophilia in the absence of other markers is controversial. The Australian treatment guide recommends monitoring for potential myocarditis through baseline and weekly (for the first month) assessments, including vitals, laboratory evaluation of peripheral eosinophils, CRP, ECG, and troponin levels. In the presence of suggestive symptoms of myocarditis, troponin more than twice the upper limit of normal or CRP over 100 mg/L is associated with 100% sensitivity. (Ref: Ronaldson KJ, Taylor AJ, Fitzgerald PB, Topliss DJ, Elsik M, McNeil JJ. Diagnostic characteristics of clozapine-induced myocarditis identified by an analysis of 38 cases and 47 controls. Journal of Clinical Psychiatry. 2010 Aug 1;71(8):976.; https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6213891/)

The correct answer is: Eosinophilia

Question **99**

Not answered

Marked out of 1.00

Flag question

Mr. Nigel Banks is a 51-year-old man with a history of depression treated with Amitriptyline for 1 year. He stopped taking his medication 3 days ago as he felt his mood has returned to its normal state for several months. Now he reports experiencing tremors, nausea and vomiting. Which of the following is most likely to account for his symptoms?

Select one:

- Serotonin syndrome
- Hypertensive crisis
- Cholinergic rebound
- Neuroleptic malignant syndrome
- Dopaminergic crisis

Your answer is incorrect.

The patient's symptoms are most likely the result of abrupt withdrawal of his Amitriptyline. Tricyclic antidepressants can cause significant symptoms as a result of cholinergic rebound after prolonged blockade by the cholinergic receptors.

The correct answer is: Cholinergic rebound

Question **100**

Not answered

Marked out of 1.00

Flag question

Tetrabenazine is a medication used to manage the symptoms of several hyperkinetic movement disorders, including Huntington's Disease, Tourette's Syndrome and Tardive Dyskinesia. Which of the following most accurately describes its mechanism of action?

Select one:

- Inhibition of vesicular monoamine transporter 2
- Dopamine agonist
- Inhibition of acetylcholine receptor muscarinic receptors
- 5HT-3A antagonist
- GABA agonist

Your answer is incorrect.

Tetrabenazine depletes monoamines such as dopamine, serotonin, noradrenaline and histamine. It

appears to achieve this by reversibly inhibiting vesicular monoamine transporter 2, reducing reuptake of monoamines into synaptic vesicles and so diminishing their availability.

The correct answer is: Inhibition of vesicular monoamine transporter 2

Question **101**

Not answered

Marked out of 1.00

Flag question

Antidepressants usually affect the function of serotonin and noradrenaline to produce their therapeutic effects. Which of the following antidepressants has the most effect on dopamine levels?

Select one:

- Paroxetine
- Duloxetine
- Sertraline
- Citalopram
- Fluoxetine

Your answer is incorrect.

The pharmacological properties of SNRIs such as duloxetine depend on their dose. Serotonergic reuptake inhibition predominates at low doses, while additional NE reuptake inhibition occurs at medium doses. At higher doses, they weakly inhibit dopamine reuptake. Recent evidence indicates that NE transporter also transports some dopamine in the prefrontal cortex. The prefrontal cortex has low concentration of dopamine reuptake transporters (DAT); therefore, SNRIs such as duloxetine can substantially increase dopaminergic neurotransmission in prefrontal cortex. Ref: Carboni, E., Tanda, G. L., Frau, R., and Di Chiara, G. (1990). Blockade of the noradrenaline carrier increases extracellular dopamine concentrations in the prefrontal cortex: evidence that dopamine is taken up in vivo by noradrenergic terminals. J. Neurochem. 55, 1067–1070.

The correct answer is: Duloxetine

Question **102**

Not answered

Marked out of 1.00

Flag question

Donepezil is an acetylcholinesterase inhibitor often used in the treatment of Alzheimer's disease. Which of the following hepatic enzyme metabolizes Donepezil?

Select one:

- CYP450 2D6
- CYP450 3A1
- CYP450 1A2
- CYP450 4A2
- CYP450 2A4

Your answer is incorrect.

Donepezil is broken down into 4 metabolites, 2 of which are active. It is metabolised by CYP 450 isoenzymes 2D6 as well as 3A4.

The correct answer is: CYP450 2D6

Question **103**

Not answered

Marked out of 1.00

Flag question

Mr. Daniel Lightfoot is a 32-year-old man with a diagnosis of bipolar affective disorder for which he has been taking Lithium for the past 5 years. He now reports feeling the urge to drink more water than usual and is urinating a large amount. Which of the following is the most likely cause?

Select one:

- Uraemia
- Nephrotic syndrome
- Lithium toxicity
- Hyperthyroidism
- Diabetes insipidus

Your answer is incorrect.

Lithium accumulates in the cells of the collecting duct of the nephrons and inhibits ADH's ability to increase water permeability. Thus, chronic lithium use affects the nephron's ability to respond to ADH. This ADH-resistance (nephrogenic diabetes insipidus) results in polydipsia and polyuria, affecting ~40% of patients on long-term lithium therapy Ref: Ellingrod VL. Lithium-induced diabetes insipidus: Prevention and management. Current Psychiatry. 2013 Jul;12(7):42-5.

The correct answer is: Diabetes insipidus

Question **104**

Not answered

Marked out of 1.00

Flag question

Mrs. Emma Premack is a 61-year-old woman who has Type II diabetes mellitus and has unfortunately developed significant pain due to peripheral neuropathy and is presenting with notable depression. Which of the following medications would be most helpful for her symptoms?

Select one:

- Citalopram
- Venlafaxine
- Mirtazapine
- Duloxetine
- Sertraline

Your answer is incorrect.

Duloxetine is licensed as an antidepressant and also for the treatment of peripheral nerve pain and urinary incontinence.

The correct answer is: Duloxetine

Question **105**

Not answered

Marked out of 1.00

Flag question

Cytochrome 450 1A2 is an important enzyme in the metabolism of a variety of medications. Which of the following induces its production?

Select one:

- Smoking
- St John's Wort

- Cimetidine
- Grapefruit juice
- Fluvoxamine

Your answer is incorrect.

Smoking cigarettes, taking modafinil or omeprazole can induce CYP450 1A2 production. The other options presented inhibit this enzyme to variable degree.

The correct answer is: Smoking

Question **106**

Not answered

Marked out of 1.00

Flag question

Antidepressants can be associated with a withdrawal syndrome when they are stopped or the dose is reduced. Which of the following drugs when stopped suddenly is most likely to be associated with withdrawal symptoms?

Select one:

- Fluoxetine
- Sertraline
- Venlafaxine
- Citalopram
- Mirtazapine

Your answer is incorrect.

Venlafaxine and paroxetine are associated with significant withdrawal syndromes and should be reduced and discontinued with caution.

The correct answer is: Venlafaxine

Question **107**

Not answered

Marked out of 1.00

Flag question

Which of the following antidepressants is least likely to cause hypotension?

Select one:

- Venlafaxine
- Lofepramine
- Trazodone
- Phenelzine
- Mirtazapine

Your answer is incorrect.

Orthostatic (postural) hypotension is relatively common with Mirtazapine and Trazodone. It is also linked to tricyclic antidepressants and older monoamine oxidase inhibitors. Postural hypotension can occur with SSRIs and SNRIs, but is less common.

The correct answer is: Venlafaxine

Question **108**

Not answered

Marked out of 1.00

Flag question

Mr. Alan Mitchell is a 63-year-old man with a history of depression and cardiac failure for which he takes Furosemide and Ramipril. He presents to his GP with leg cramps, progressive weakness and constipation. An ECG in the surgery demonstrated T wave flattening. Which of the following electrolyte imbalances is most likely to account for his symptoms?

Select one:

- Hypocalcaemia
- Hyponatraemia
- Hypokalaemia
- Hypercalcaemia
- Hyperkalaemia

Your answer is incorrect.

T wave flattening is characteristic of hypokalaemia. It often presents with non-specific leg cramps and weakness before progressing to paresis and a variety of renal, cardiac, neurological and gastrointestinal disturbances.

The correct answer is: Hypokalaemia

Question **109**

Not answered

Marked out of 1.00

Flag question

Benzodiazepines are often used in clinical practice as short-term sedatives and hypnotics. Which of the following best describes their mechanism of action?

Select one:

- Decreases neuronal excitability via GABAA receptor
- Decreases neuronal excitability via GABAB receptor
- Increases neuronal excitability via GABAA receptor
- Increases neuronal excitability via GABAB receptor
- Increases neuronal excitability via GABAC receptor

Your answer is incorrect.

Benzodiazepines increase the efficiency of GABA by hyperpolarising the cell membrane of neurons after attaching to the GABAA receptor subtype between the α and γ subunits. This decreases the excitability of the neuron and so the brain overall.

The correct answer is: Decreases neuronal excitability via GABAA receptor

Question **110**

Not answered

Marked out of 1.00

Miss Annunziata Farage is a 25-year-old woman who has been brought to her local A&E after collapsing and making odd movements in the street. The medical staff are querying whether the episode was epileptic in nature or if it was psychologically driven. Which of the following would be most specific for differentiating between epileptic and pseudo seizures in this case?

Flag question

Select one:

- Neurological examination
- Presence or absence of incontinence
- Prolactin level
- History from an observer
- EEG

Your answer is incorrect.

Prolactin levels taken 10-20 minutes after a possible seizure are highly specific (around 96%) for identifying epileptic seizures. (Ref: Chen DK, So YT, Fisher RS. Use of serum prolactin in diagnosing epileptic seizures: report of the Therapeutics and Technology Assessment Subcommittee of the American Academy of Neurology. Neurology; 2005 Sep 13;65(5):668-75).

The correct answer is: Prolactin level

Question **111**

Not answered

Marked out of 1.00

Flag question

Mrs. Ivy Richards is a 76-year-old woman who is due to be started on memantine. Her daughter has some queries about its action and use. Which of the following is not true regarding memantine?

Select one:

- It can cause bradycardia as a side-effect
- It is indicated in moderate to severe vascular dementia
- Its elimination half-life is 60-80 hours.
- It potentiates the effect of Acetylcholine at nicotinic receptors.
- It is an antagonist at NMDA receptors

Your answer is incorrect.

Memantine is a NMDA antagonist used in moderate to severe Alzheimer's disease. Its half-life is 60-80 hours. Common side effects are hallucinations and confusion. Acetylcholinesterase inhibitors, not memantine, are associated with bradycardia.

The correct answer is: It can cause bradycardia as a side-effect

Question **112**

Not answered

Marked out of 1.00

Flag question

Mr Evan Stevens is a 32-year-old man with a history of treatment resistant schizophrenia who will be started on clozapine in the next few weeks. When he attended the clozapine clinic for an information session, he was advised about the effect of smoking on Clozapine. Which of the following is most likely to be seen in smokers on clozapine?

Select one:

- Increased risk of neutropenia
- Increased serum Clozapine levels
- Decreased urge to smoke
- Increased risk of paralytic ileus

- Decreased serum Clozapine levels

Your answer is incorrect.

Smoking stimulates CYP450 1A2, which metabolises clozapine. This results in lower serum levels.

The correct answer is: Decreased serum Clozapine levels

Question **113**

Not answered

Marked out of 1.00

Flag question

Ingested psychotropic medications are metabolised in a variety of ways. Which of the following drugs follows zero order kinetics?

Select one:

- Vortioxetine
- Lithium
- Sertraline
- Amisulpride
- Phenytoin

Your answer is incorrect.

In zero order kinetics, a constant amount, rather than percentage, of the drug is eliminated per unit of time even when therapeutic doses are taken, e.g. a fixed 1 mg per hour. This is independent of the concentration of the drug in the serum. Examples of drugs following zero order kinetics are phenytoin, aspirin and ethanol.

The correct answer is: Phenytoin

Question **114**

Not answered

Marked out of 1.00

Flag question

Lithium's efficacy depends upon it remaining within a therapeutic window in terms of serum level. Its serum level can be significantly affected by certain medications. Which of the following drugs reduces serum lithium levels?

Select one:

- Theophylline
- Furosemide
- Lisinopril
- Haloperidol
- Tramadol

Your answer is incorrect.

Drugs that reduce lithium levels include theophylline, caffeine, and acetazolamide. Additionally, increasing dietary sodium intake may also reduce lithium levels by prompting the kidneys to excrete more lithium. Furosemide, thiazides, ibuprofen and ACE inhibitors tend to increase lithium levels. Tramadol and certain other opiates can precipitate serotonin syndrome in association with lithium. Haloperidol and other antipsychotics can increase the risk of toxic encephalopathy, without affecting

lithium levels.

The correct answer is: Theophylline

Question **115**

Not answered

Marked out of 1.00

Flag question

Agomelatine is a relatively new antidepressant, which binds to melatonin receptors where it acts as an agonist. Which of the following other receptors does agomelatine bind to?

Select one:

- 5-HT3A
- 5-HT1A
- 5-HT2B
- 5-HT2C
- 5-HT5A

Your answer is incorrect.

Agomelatine is a potent agonist at melatonin MT1 and MT2 receptors and an antagonist at 5-HT2C receptors. The psychotropic effects of agomelatine are likely related to both melatonergic and 5-HT effects. (Ref: https://www.ncbi.nlm.nih.gov/pubmed/20946699)

The correct answer is: 5-HT2C

Question **116**

Not answered

Marked out of 1.00

Flag question

Social anxiety is characterised by avoidance of and anxiety over social situations where an individual will be observed by others. Which of the following drugs has proven effective for social anxiety?

Select one:

- Pregabalin
- Quetiapine
- Carbamazepine
- Lamotrigine
- Risperidone

Your answer is incorrect.

Some research has found Pregabalin to be effective in the treatment of social anxiety. Pregabalin 600 mg/day was an effective and well-tolerated treatment of social anxiety disorder. Somnolence and dizziness were the most frequently reported adverse events. Please note that pregabalin is not endorsed by NICE which instead recommends SSRIs, Venlafaxine and MAOIs, particularly moclobemide. (Ref: Pande AC et al. Efficacy of the novel anxiolytic pregabalin in social anxiety disorder: a placebo-controlled, multicenter study. Journal Clinical Psychopharmacology, 2004 Apr; 2:141-9).

The correct answer is: Pregabalin

Question **117**

Not answered

The CATIE study showed what percentage risk of metabolic syndrome in patients with schizophrenia receiving antipsychotic treatment?

Marked out of 1.00

Flag question

Select one:

- 60 to 65%
- 50 to 55%
- 30 to 35%
- 20 to 25%
- 40 to 45%

Your answer is incorrect.

The prevalence of metabolic syndrome in CATIE cohort was 40.9% and 42.7%, respectively using the NCEP and AHA derived criteria. In females it was 51.6% and 54.2% using the NCEP and AHA criteria, compared to 36.0% (p = .0002) and 36.6% (p = .0003), respectively for males. Ref; McEvoy JP, et al. Schizophrenia research. 2005 Dec 1;80(1):19-32.

The correct answer is: 40 to 45%

Question **118**

Not answered

Marked out of 1.00

Flag question

Which of the following is not a prodrug?

Select one:

- Lisdexamphetamine
- Fosphenytoin
- Oxcarbazepine
- Prednisone
- Desipramine

Your answer is incorrect.

A prodrug is a medication or compound that, after administration, is metabolised into a pharmacologically active drug. Desipramine is the metabolite of imipramine. All the others listed above are pro-drugs.

The correct answer is: Desipramine

Question **119**

Not answered

Marked out of 1.00

Flag question

Mr Nicholas Orgreave is a 31-year-old man with a history of paranoid schizophrenia who has recently been started on a Haloperidol depot after being admitted following a relapse due to medication non-compliance in the community. He becomes agitated on the ward in response to auditory hallucinations and the ward staff request advice around tranquilisation. Which of the following would be most appropriate to give him as a first-line option?

Select one:

- Oral Diazepam 5mg
- Oral Chlorpromazine 100mg
- Oral Lorazepam 2mg
- Inhaled Loxapine

- Oral Olanzapine 5mg

Your answer is incorrect.

Second generation antipsychotics are generally preferred in the treatment of agitation related to psychotic illness. Benzodiazepines can be effective in sedating a patient but do not address the underlying drivers for the behaviour i.e. psychosis in this case.

The correct answer is: Oral Olanzapine 5mg

Question **120**

Not answered

Marked out of 1.00

Flag question

What is the annual risk of developing tardive dyskinesia in individuals in individual exposed to long-term typical antipsychotics?

Select one:

- 25-30%
- 5-10%
- 20-25%
- 15-20%
- 30-35%

Your answer is incorrect.

Tardive dyskinesia is a disorder that results in involuntary, repetitive body movements. This may include grimacing, sticking out the tongue, or smacking the lips. It is associated with long-term antipsychotic use. Long term use of typical antipsychotics, such as Haloperidol, are associated with a risk of around 25-30% of developing tardive dyskinesia, at a rate of 5% of patients per year of antipsychotic exposure. Nearly 50% of these cases are reversible. Ref: Maudsley Prescribing Guidelines 13th edition. Also see Caligiuri M. Tardive dyskinesia: a task force report of the American Psychiatric Association. Hosp Community Psychiatry 1993; 44:190.

The correct answer is: 5-10%

Question **121**

Not answered

Marked out of 1.00

Flag question

Mr Adrian Dawes is a 48-year-old man who is experiencing a significant depressive episode with marked difficulty getting to sleep and then waking frequently in the night. He is finding this exhausting and has asked about medication that might help him get to and stay asleep. Which of the following would be most appropriate for him?

Select one:

- Lorazepam
- Diazepam
- Nitrazepam
- Zopiclone
- Temazepam

Your answer is incorrect.

Temazepam is recommended as a medication to aid the initiation and maintenance of sleep. It has a mean half-life of around 8 hours, although the range can be 8-20 hours. This usually limits early morning tiredness.

The correct answer is: Temazepam

Question **122**

Not answered

Marked out of 1.00

Flag question

Mr Ian Watkins is a 37-year-old man with a long-standing history of bipolar affective disorder that has not responded well to medication trials. He has read about Topiramate as a potential treatment option and would want to know more about it. Which of the following is a common side effect of Topiramate?

Select one:

- Increased appetite
- Visual disturbance
- Weight loss
- Derealisation
- Increased sleep

Your answer is incorrect.

Weight loss is a common side effect of Topiramate.

The correct answer is: Weight loss

Question **123**

Not answered

Marked out of 1.00

Flag question

Lithium is a very effective mood stabiliser but needs to be maintained within a therapeutic range, with a risk of toxicity if the serum level is too high. Which of the following is most likely to make lithium toxicity worse?

Select one:

- Paracetamol
- Furosemide
- Ibuprofen
- Prednisolone
- Theophylline

Your answer is incorrect.

Furosemide significantly increases the serum level of Lithium and can also increase the risk of hypokalaemia and torsades de pointes. Ibuprofen and other NSAIDs can increase the serum Lithium level although the risk varies within the drug class. Prednisolone can increase the risk of hypokalaemia. Theophylline can decrease the level of Lithium.

The correct answer is: Furosemide

Question **124**

Not answered

Marked out of 1.00

Flag question

GHB (γ-Hydroxybutyric acid) is a naturally occurring neurotransmitter and a psychoactive drug. It is a precursor to GABA, glutamate, and glycine in certain brain areas. Which of the following receptors does it work on?

Select one:

- GABA-B
- GABA-A
- Glutamate
- Glycine
- Calcitonin

Your answer is incorrect.

GHB functions at both the GABA-B receptor but also at the γ-hydroxybutyrate (GHB) receptor (GHBR), which is G-protein-coupled. GHBR is also a transporter for riboflavin. Ref: Crunelli V, Emri Z, Leresche N. Unravelling the brain targets of γ-hydroxybutyric acid. Current opinion in pharmacology. 2006 Feb 1;6(1):44-52.

The correct answer is: GABA-B

Question **125**

Not answered

Marked out of 1.00

Flag question

Diazepam is a commonly used medication. Which of the following receptors does it act on?

Select one:

- M1
- Both GABA-A and GABA-B
- M2
- GABA-A
- GABA-B

Your answer is incorrect.

Diazepam binds to the GABA-A receptors, which are ligand-gated chloride-selective ion channels, which are activated by GABA. Diazepam and other benzodiazepines enhance the binding of GABA, which leads to an increase in chloride within the neuron that causes hyperpolarization. This makes neuronal excitation more difficult and reduces the likelihood of neuronal firing.

The correct answer is: GABA-A

Question **126**

Not answered

Marked out of 1.00

Flag question

New medications must undergo testing prior to being approved for general use to ensure their efficacy and safety. How many stages of approval must a medication pass through prior to general use?

Select one:

- 2
- 5
- 3

- 4
- 6

Your answer is incorrect.

There are 6 stages of clinical trials related to a new medication, with the last stage (phase 4) comes after general use begins (post-marketing). The stages are: Preclinical - Animal or in vitro studies to determine toxicity, pharmacokinetics and initial indications of efficacy. Phase 0 - Small sample of human subjects exposed to low dose to further investigate pharmacokinetics . Phase 1 - Larger sample of healthy volunteers with increasing doses used to determine appropriate ranges. Phase 2 - Testing on group of patients to determine efficacy in clinical practice Phase 3 - Testing on larger group of patients to establish efficacy, effectiveness and safety Phase 4 - Post-marketing monitoring after the medication is released for general use.

The correct answer is: 5

Question **127**

Not answered

Marked out of 1.00

Flag question

Mr Johnson Nash is a 57-year-old man with a diagnosis of paranoid schizophrenia who has been well on the same medication for several years. Unfortunately he has now developed autoimmune hepatitis, which his hepatology team is struggling to manage. There are concerns about whether he can continue on his current medication regime for schizophrenia. Which of the following antipsychotics are most appropriate for individuals with significant hepatic impairment?

Select one:

- Fluphenazine
- Lurasidone
- Amisulpride
- Quetiapine
- Chlorpromazine

Your answer is incorrect.

Amisulpride and Sulpiride are predominantly renally excreted and both are safer antipsychotic medicines that could be used in patients with hepatic impairmemt. (Ref: Telles-Correia D, et al. Psychotropic drugs and liver disease: A critical review of pharmacokinetics and liver toxicity. World Journal of Gastrointestinal Pharmacology and Therapeutics. 2017;8(1):26Å38)

The correct answer is: Amisulpride

Question **128**

Not answered

Marked out of 1.00

Flag question

Which of the following SSRIs has an active metabolite?

Select one:

- Sertraline
- Fluvoxamine
- Escitalopram
- Paroxetine
- Citalopram

Your answer is incorrect.

Sertraline also has an active metabolite, N-desmethyl sertraline, which is 10 times less potent than sertraline itself. Fluoxetine also has an active metabolite norfluoxetine with a long half-life.

The correct answer is: Sertraline

Question **129**

Not answered

Marked out of 1.00

Flag question

Miss Janet Baldwin is a 24-year-old woman with a new diagnosis of first episode psychosis. She has a history of recurrent hypotension without an established cause and is keen that any medication does not worsen this. Which of the following antipsychotics is least likely to cause hypotension?

Select one:

- Chlorpromazine
- Risperidone
- Aripiprazole
- Haloperidol
- Clozapine

Your answer is incorrect.

Aripiprazole is associated with a low risk of postural hypotension compared to other antipsychotics. It is thought to be linked to antagonism of alpha-1-adrenergic receptors which are associated with vasoconstriction. (Ref: Polcwiartek C, et al. Cardiovascular safety of antipsychotics: a clinical overview, Expert Opinion on Drug Safety; 2016;15:5:679-688).

The correct answer is: Aripiprazole

Question **130**

Not answered

Marked out of 1.00

Flag question

The half-lives of psychiatric medications vary widely and this can influence the time necessary for them to reach steady state. Which of the following medications has a half-life of more than 50 hours if taken orally?

Select one:

- Citalopram
- Flupenthixol
- Mirtazapine
- Venlafaxine
- Aripiprazole

Your answer is incorrect.

Aripiprazole has a half-life of around 75 hours. Flupentixol and Citalopram have half-lives of around 35 hours. Venlafaxine has a half-life of 3 to 7 hours, while its active metabolite, desvenlafaxine, has a half-life of 9 to 13 hours. Mirtazapine has a half-life of 20-40 hours.

The correct answer is: Aripiprazole

Question **131**

Not answered

Marked out of 1.00

Flag question

Many psychotropic medications can reduce the seizure threshold as a side-effect. Which of the following antipsychotics is associated with the highest risk of seizures?

Select one:

- Risperidone
- Clozapine
- Aripiprazole
- Haloperidol
- Fluphenazine

Your answer is incorrect.

Clozapine is the highest risk antipsychotic for seizures, with a cumulative risk 10% for seizure activity. Chlorpromazine is the highest risk first generation antipsychotic for seizures. (Ref: Bhatti M, Dorriz P and Mehndiratta P. Impact of Psychotropic drugs on Seizure threshold. Neurology;2017 April;88(16 Supplement).

The correct answer is: Clozapine

Question **132**

Not answered

Marked out of 1.00

Flag question

MAOI antidepressants have been largely replaced by other agents due to concerns around significant interactions with other medications and foods. Which of the following should not be used concurrently with a MAOI?

Select one:

- Decongestants
- Antihistamines
- Anticonvulsants
- ACE inhibitors
- Penicillins

Your answer is incorrect.

MAOIs have been associated with hypertensive crises when used with over the counter decongestants. (Ref: Carlo AD, Alpert JE. Catastrophic Drug-Drug Interactions in Psychopharmacology. Psychiatric Annals. 2016 Aug 12;46(8):439-47.

The correct answer is: Decongestants

Question **133**

Not answered

Marked out of 1.00

Flag

Medications within the same class can have widely ranging half-lives, which can influence their potential for adverse effects and dosing regimes. Which of the following SSRI antidepressants has the shortest half-life?

Select one:

question

- Fluoxetine
- Fluvoxamine
- Citalopram
- Paroxetine
- Sertraline

Your answer is incorrect.

Fluvoxamine is thought to have a half-life of around 16 hours. The half-lives of other SSRIs are Paroxetine - 21 hours, Sertraline - 26 hours, Citalopram - 35 hours, Fluoxetine - 1-4 days (Norfluoxetine - 7-9 days). (Ref: Marazziti D, Avella MT, Basile L, Mucci F, Dell'Osso L. Pharmacokinetics of serotonergic drugs: focus on OCD. Expert Opinion on Drug Metabolism & Toxicology. 2019 Apr 3;15(4):261-73).

The correct answer is: Fluvoxamine

Question **134**

Not answered

Marked out of 1.00

Flag question

Lurasidone is a relatively new antipsychotic medication. Which of the following is true regarding its absorption?

Select one:

- No effect on food
- Used sublingually
- Food increases absorption
- Administered nasally
- Food decreases absorption

Your answer is incorrect.

Lurasidone is weakly soluble in water; its bioavailability is limited after regular oral ingestion. When administered with a meal of at least 350 calories, the bioavailability increases by 2-3 times. (Ref: Jaeschke RR, et al. Lurasidone: The 2016 update on the pharmacology, efficacy and safety profile. Pharmacological Reports. 2016 Jul 1;68(4):748-55.).

The correct answer is: Food increases absorption

Question **135**

Not answered

Marked out of 1.00

Flag question

Zolpidem is a commonly used hypnotic medication that has a shorter half-life than Zopiclone. Which of the following receptors does it act upon?

Select one:

- GABA-A
- Kappa
- Mu
- GABA-B
- GABA-A and -B

Your answer is incorrect.

Zolpidem acts at the GABA-A receptor benzodiazepine site; but unlike benzodiazepines that non-selectively bind to all GABA-A receptor subtypes, zolpidem exhibits preferential binding with a high affinity for the alpha1 subunit, with lower affinity for alpha2/3 subunits. Its sedative effects (and anti-convulsant properties in vitro) are derived exclusively from its action on the alpha1 subunit. (Ref: Neumann E, et al. Zolpidem Activation of Alpha 1-Containing GABAA Receptors Selectively Inhibits High Frequency Action Potential Firing of Cortical Neurons. Front Pharmacol. 2018;9:1523.).

The correct answer is: GABA-A

Question **136**

Not answered

Marked out of 1.00

Flag question

Miss Jane Sessions is a 28-year-old woman with a diagnosis of first episode psychosis. She has agreed to start an antipsychotic but is worried as she has heard that they are associated with weight gain. By which of the following mechanisms can antipsychotics lead to an increase in weight?

Select one:

- 5HT2C agonism
- H1 antagonism
- 5HT1A antagonism
- M1 antagonists
- M3 agonists

Your answer is incorrect.

Weight gain associated with antipsychotics is thought to be linked to multiple mechanisms, including antagonism at H1 receptors, which increases appetite by impairing hypothalamic functioning and falsely indicating negative energy balance. It is also associated with 5HT2C antagonism, which delays the onset of satiety, and 5HT1A agonism, increasing food intake. Interesting, dopamine blockade may also increase weight by impairing limbic systems and leading to more reward-seeking behaviour, such as eating. (Ref: He M, et al. The role of hypothalamic H1 receptor antagonism in antipsychotic-induced weight gain. CNS drugs. 2013 Jun 1;27(6):423-34).

The correct answer is: H1 antagonism

Question **137**

Not answered

Marked out of 1.00

Flag question

Old age is associated with significant physiological changes that impact upon the effectiveness and tolerance of medications. With respect to drug distribution, which of the following tends to occur with normal aging?

Select one:

- Increase in lipophilic drug distribution and decrease in hydrophilic drug distribution
- Increase in both lipophilic and hydrophilic drug distribution
- Decrease in both hydrophilic and lipophilic drug distribution.
- Decrease in hydrophilic drug distribution and no change in lipophilic drug distribution
- Increase is hydrophilic drug distribution and decrease in lipophilic drug distribution

Your answer is incorrect.

Older people tend to have a higher volume of distribution for lipophilic drugs and a lower one for hydrophilic medications. They tend to have reduced effectiveness of first pass hepatic metabolism and CYP isoform activity. Renal clearance is often reduced. (Ref: Reeve E, Trenaman SC, Rockwood K, Hilmer SN. Pharmacokinetic and pharmacodynamic alterations in older people with dementia. Expert opinion on drug metabolism & toxicology. 2017 Jun 3;13(6):651-68).

The correct answer is: Increase in lipophilic drug distribution and decrease in hydrophilic drug distribution

Question **138**

Not answered

Marked out of 1.00

Flag question

Pregabalin is an antiepileptic medication that is also used in chronic pain and anxiety. Which of the following is true regarding this medication?

Select one:

- It has poor oral bioavailability
- It has no association with weight gain
- It is excreted without being metabolized
- Its half-life is around 24 hours
- It is an agonist of calcium channels

Your answer is incorrect.

Pregabalin undergoes very little metabolism and around 98% is unchanged and excreted renally. It has good oral bioavailability. Its half-life is around 6-11 hours so is usually administered 2-3 times per day. It is an inhibitor of certain calcium channels. It is associated with some increase in weight. Ref: Chincholkar M. Gabapentinoids: pharmacokinetics , pharmacodynamics and considerations for clinical practice. British Journal of Pain. 2020 May;14(2):104-14

The correct answer is: It is excreted without being metabolized

Question **139**

Not answered

Marked out of 1.00

Flag question

Miss Susan Stolet is a 24-year-old woman who has recently been diagnosed with epilepsy and started on an anticonvulsant. She returns to clinic with her mother who has noticed that her behaviour has changed and become more aggressive since the medication was started. Which of the following drugs is most likely to be responsible?

Select one:

- Topiramate
- Sodium Valproate
- Lamotrigine
- Carbamazepine
- Levetiracetam

Your answer is incorrect.

Levetiracetam is associated with a high rate of behavioural side-effects, with one study suggesting that 13% of patients may experience this. Topiramate is also associated with a high rate of behavioural and

aggressive side-effects, although this can be associated with reduced seizure control, which may indicate it is partially an ictal phenomenon. (Ref: Brodie MJ, Besag F, Ettinger AB, et al. Epilepsy, Antiepileptic Drugs, and Aggression: An Evidence-Based Review. Pharmacology Review; 2016;68(3):563Å602).

The correct answer is: Levetiracetam

Question **140**

Not answered

Marked out of 1.00

Flag question

A GP has started a psychiatric patient on Ramipril for treatment of his hypertension. The GP is worried about drug interactions as he is on a number of psychiatric medicines. Which of the following drugs can have elevated levels if prescribed with Ramipril?

Select one:

- Olanzapine
- Lithium
- Risperidone
- Mirtazapine
- Agomelatine

Your answer is incorrect.

Ramipril is an ACE inhibitor. Lithium is known to have significant drug interactions with medications that affect kidney tubule functioning, including NSAIDs, diuretics and ACE inhibitors. These medications generally result in a significant increase in serum Lithium levels and can lead to toxicity. (Ref: Hommers L, et al. The combination of lithium and ACE inhibitors: Hazardous, critical, possible?. Clinical drug investigation. 2019 May 1;39(5):485-90).

The correct answer is: Lithium

Question **141**

Not answered

Marked out of 1.00

Flag question

Protein binding of medications can have a significant effect upon their efficacy and metabolism. Which of the following antidepressants is most highly protein bound?

Select one:

- Sertraline
- Paroxetine
- Escitalopram
- Fluoxetine
- Citalopram

Your answer is incorrect.

Sertraline is 98% protein bound. The percentages for other SSRIs are Citalopram - 80%, Escitalopram - 56%, Fluoxetine - 95% and Paroxetine - 93-95% (Ref: Ciraulo, DA; Shader, RI, eds. (2011). Pharmacotherapy of Depression. SpringerLink (2nd ed.). New York, NY: Humana Press).

The correct answer is: Sertraline

Question **142**

Not answered

Marked out of 1.00

Flag question

Antidepressants can be associated with QTc prolongation and potential torsades de pointes. Which of the following antidepressants is least associated with QTc prolongation?

Select one:

- Mirtazapine
- Venlafaxine
- Citalopram
- Imipramine
- Moclobemide

Your answer is incorrect.

Mirtazapine, Sertraline and Reboxetine are least associated with QTc prolongation at therapeutic doses. Tricyclics are best avoided completely in patients at risk of serious arrhythmias. Citalopram, Moclobemide and venlafaxine should be used with caution or avoided in those at risk of serious arrhythmias. (Ref: Rochester MP, et al. Evaluating the risk of QTc prolongation associated with antidepressant use in older adults: a review of the evidence. Therapeutic advances in drug safety. 2018 Jun;9(6):297-308.).

The correct answer is: Mirtazapine

Question **143**

Not answered

Marked out of 1.00

Flag question

Mr. Thomas Eaton is a 25-year-old man with a diagnosis of paranoid schizophrenia and depression. He has recently been started on haloperidol and paroxetine but is brought into his local A&E by his family after developing a fever and jerking in his limbs. In hospital, it is found that his pulse and blood pressure are elevated and he is demonstrating clonus. Which of the following is he most likely to be experiencing?

Select one:

- Acute dystonia
- Neuroleptic malignant syndrome
- Serotonin syndrome
- Cholinergic toxicity
- Malignant hyperthermia

Your answer is incorrect.

This is most likely to represent serotonin syndrome, which is characterised by elevated bodily temperature, hypertension, tachycardia, dilated pupils, myoclonus, clonus and agitation. It tends to be acute in onset. Neuroleptic malignant syndrome can present with similar symptoms but is less acute in onset and is not associated with myoclonus or clonus.

The correct answer is: Serotonin syndrome

Question **144**

Not answered

John is a 59-year-old man taking psychotropic medications for a long time. He has recently been diagnosed with interstitial nephritis. Which of the following can cause this adverse reaction?

Marked out of 1.00

Flag question

Select one:

- Carbamazepine
- Haloperidol
- Lithium
- Fluoxetine
- Risperidone

Your answer is incorrect.

Long term Lithium use is associated with chronic tubulointerstitial nephropathy, which is characterised by declining GFR without marked proteinuria. It is also associated with long-term thyroid and parathyroid dysfunction. (Ref: Kovvuru K, Kanduri SR, Thongprayoon C, Cheungpasitporn W. Lithium and nephrotoxicity: Nephrology's perspectives. Bipolar Disorders. 2020 Jun;22(4):331-3.).

The correct answer is: Lithium

Question **145**

Not answered

Marked out of 1.00

Flag question

Which of the following medications acts as a D2 receptor agonist?

Select one:

- Haloperidol
- Clozapine
- Aripiprazole
- Lurasidone
- Sulpride

Your answer is incorrect.

Aripiprazole acts as a partial agonist at D2 receptors making it unique amongst antipsychotic medications. None of the other listed medications have agonistic actions at D2.

The correct answer is: Aripiprazole

Question **146**

Not answered

Marked out of 5.00

Midcingulate Cortex
Anterior Insula
Basolateral amygdala
Globus Pallidus Externa
Globus Pallidus Interna
Periaqueductal Grey
Sensorimotor cortex
Subcallosal Cingulate Cortex
Subthalamic Nucleus
Ventral Internal Capsule
Ventral Tegmental Area
Ventromedial Thalamus
Visual cortex

Theme- Deep Brain Stimulation Targets
Lead In: For each of the following conditions, choose the target regions from the above list that provide therapeutic relief when stimulated using DBS.

1. Parkinson's Disease (Choose Two)

2. Treatment Resistant OCD (Choose Two)

3. Treatment Resistant Depression (Choose ONE)

Explanation:
• DBS at the subthalamic nucleus and the internal globus pallidus is endorsed by many guidelines for treatment of Parkinson's disease. It may be as effective as medications, but without motor side effects in many patients.
• DBS of the internal capsule has shown positive effects for OCD. STN is another target of interest for OCD.
• DBS of ventromedial thalamus can also reduce abnormal movements in patients with Tourette's disorder.
• DBS of subcallosal cingulate (Cg25 area) reduces depression in treatment resistant cases Casagrande SC, et al. Deep brain stimulation in Tourette's syndrome: evidence to date. Neuropsychiatric disease and treatment. 2019;15:1061.
Crowell AL, et al. Long-term outcomes of subcallosal cingulate deep brain stimulation for treatment-resistant depression. American Journal of Psychiatry. 2019 Nov 1;176(11):949-56.

Question **147**

Not answered

Marked out of 7.00

Aromatase knockout
BDNF haploinsufficient mice
Chronic restraint
Circadian rhythm disruption
Dysbindin knockout
ERK1 knockout
Frontal ablation
Maternal Immune Activation
Maternal Separation
Neonatal clomipramine
Neonatal ventral hippocampal lesion
Olfactory bulbectomy
Social Defeat

Theme- Animal Models of Psychiatric Disorders II
Lead In: For each of the following conditions, choose the appropriate animal model from the above list

1. Bipolar Disorder (Choose THREE)

2. Depressive Disorder (Choose FOUR)

Disrupted circadian rhythm in animal models increases sensitivity for bipolar disorder. The ClockΔ19 mutant mouse has a disrupted molecular circadian clock, exhibits bipolar-like behaviour; restoration of the circadian clock to the VTA, restores dopaminergic regulation, ameliorates the mania-like behaviors. BD-relevant behaviour is also seen in knockouts of the extracellular-signal-regulated kinase (ERK) gene that resets the master pacemaker in the suprachiasmatic nucleus via photic input. BDNF haploinsufficient mice indeed exhibit mania-like behavior, including hyperactivity, increased aggressive behavior, and appetite. The olfactory bulbectomized (OB) rat shows nocturnal hyperactivity, deficits in memory, passive avoidance behaviour and changes in food motivated and conditioned taste aversion behaviour, reminiscent of depression. Social defeat, maternal deprivation as well as chronic restraint can increase cortisol output and reproduce features of depression.
1. Ozcan H, Aydin N, Aydin MD, Oral E, Gündogdu C, Sipal S, Halici Z. Olfactory bulbectomy and raphe nucleus relationship: a new vision for well-known depression model. Nordic Journal of Psychiatry. 2020 Apr 2;74(3):194-200.
2. Beyer DK, Freund N. Animal models for bipolar disorder: from bedside to the cage. International journal of bipolar disorders. 2017 Dec 1;5(1):35.
3. Walker WH, Walton JC, DeVries AC, Nelson RJ. Circadian rhythm disruption and mental health.

Translational Psychiatry. 2020 Jan 23;10(1):1-3.

Question **148**

Not answered

Marked out of 5.00

Activated microglia
Anterior insula activation
Caudate hypermetabolism
Cerebellar hypofunction
Enhanced oxytocin release
Hippocampal hypertrophy
Reduced CSF 5-HIAA
Tryptophan Hydroxylase polymorphism

Theme- Neurobiology of Complex Behaviours
Lead In: For each of the following conditions, choose the appropriate neurobiological correlate from the above list

1. Suicidality (Choose THREE)

2. Attachment Behaviour (Choose ONE)

3. Substance Use Disorders (Choose ONE)

Inflammatory markers including cytokine excess and activation of microglia is linked to suicidality. Serotonin pathway disruptions including higher concentration of CSF 5-HIAA and tryptophan hydroxylase polymorphisms are reported in relation to suicidality.
Anterior insula is especially involved in craving for addictive substance. Damage to the anterior insula leads to an abrupt disruption of nicotine addiction.
Oxytocin plays a role in mother-infant bonding as well as affiliative behavior in both sexes. In particular, oxytocin increases gaze fixation on human faces, enhances interpersonal trust and the accuracy of empathy.
Maltese F, Papaleo F. Oxytocin Discrepancies in Social Dynamics. Neuron. 2020 Aug 19;107(4):591-3.

Question **149**

Not answered

Marked out of 4.00

5 hours
3 days
12 hours
4-5 days
7-9 days
9 hours
24 hours
18 hours

Theme- Half-lives
Lead In: Which of the following periods of time best reflect the half-lives of each of the below medications?

1. Venlafaxine

2. Aripiprazole

3. Fluoxetine

4. Sertraline

Explanation: Venlafaxine has a half-life of 5 +/-2 hours. Aripiprazole's half-life is 75 hours. The half-life of fluoxetine is 4-6 days when taken long-term and its active metabolite, norfluoxetine has a half-life of 7 to 15 days.The half-life of Sertraline is 23-26 hours.

Finish review

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